School of Computing and Augmented Intelligence
ASU IRA A. FULTON SCHOOLS OF engineering

*CSE 446 / 598*
*Software Integration and Engineering*

# Unit 2
# Software Architecture and Integration

## Lecture 2-2
## Enterprise Architecture and Business Process

Dr. Y. Chen
https://myasucourses.asu.edu/

# Unit 2 Outline



Unit 2-1 Advanced Web App Architecture

**Unit 2-2 Enterprise Architecture and Business Process**

Unit 2-3 Workflow Concepts and Workflow Foundation

Unit 2-4 Software Integration Using Power Platform

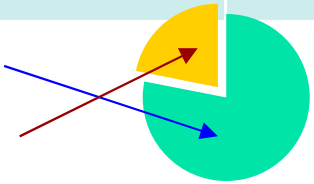
# Why to be a Software Architect/Engineer?

**Software Engineers and CS Occupations (There are many more) in U.S. DoL Occupational Outlook Handbook**

| Occupational title | Employment in 2008 | Employment in 2018 | Change in number | Change in percentage |
|---|---|---|---|---|
| Software engineers / Software Architect | 909,600 | 1,204,800 | 295,200 | 32 |
| **Computer systems analysts** | 532,200 | 640,300 | 108,100 | 20 |
| **Computer network, systems, and database administrators** | 961,200 | 1,247,800 | 286,600 | 30 |
| Computer programmers | 426,700 | 414,400 | -12,300 | -3 |
| Computer support specialists | 565,700 | 643,700 | 78,000 | 14 |
| **Computer and information systems managers** | 293,000 | 342,500 | 49,500 | 17 |

http://www.bls.gov/ooh/

IT Jobs: 4,875,600.00 (78%)
All other eng. jobs: 1,368,200.00 (22%)



# Computer and IT Jobs 2021 - 2031

## U.S. DoL Occupational Outlook Handbook http://www.bls.gov/ooh/

| Occupational title | Employment in 2021 | Employment in 2031 | Change in number | Change in percentage |
|---|---|---|---|---|
| Computer and Information Research Scientists | 33,500 | 40,600 | 7,100 | 21% |
| Computer Network Architects | 174,800 | 182,300 | 7,500 | 4% |
| Computer programmers | 174,400 | 156,600 | -17,800 | -10% |
| Computer support specialists | 875,700 | 932,100 | 56,400 | 6% |
| Computer Systems Analysts | 538,800 | 589,700 | 50,900 | 9% |
| Database Admins and Architects | 144,500 | 157,300 | 12,800 | 9% |
| Information Security Analysts | 163,000 | 219,500 | 56,500 | 35% |
| Software Developers, QA Analysts, and Testers | 1,622,200 | 2,033,600 | 411,400 | 25% |
| Web Developers & Digital Designers | 197,100 | 242,500 | 45,400 | 23% |

# 100 Best Jobs US News 2021

https://money.usnews.com/careers/best-jobs/rankings/the-100-best-jobs

| Job Title | Job # | Median Salary | Degree needed |
|---|---|---|---|
| 1. Physician Assistant | 39,300 | 112,260 | Master's |
| 2. Software Developer | 316,000 | 107,510 | Bachelor's |
| 3. Nurse Practitioner | 110,700 | 109,820 | Master's |
| 4. Medical and Health Services Manager | 133,200 | 100,980 | Bachelor's |
| 5. Physician | 18,500 | 206,500 | Doctorate |
| 6. Statistician | 14,800 | 91,160 | Master's |
| 7. Speech-Language Pathologist | 40,500 | 79,120 | Master's |
| 8. Data Scientist | 10,300 | 94,280 | Bachelor's |
| 9. Dentist | 3,700 | 155,600 | Doctorate |
| 10. Veterinarian | 14,200 | 95,460 | Doctorate |
| 11. Orthodontist | 200 | 208,000 | Doctorate |
| 12. IT Manager | 48,100 | 146,360 | Bachelor's |
| **Summary: Medical related jobs** | **280,500** | | **Master/Doctor** |
| **Summary: Computing related jobs** | **389,200** | **x 1.39** | **Bachelor** |

# 100 Best Jobs US News 2023

https://money.usnews.com/careers/best-jobs/rankings/the-100-best-jobs

| Job Title | Job # | Median Salary | Degree needed |
|---|---|---|---|
| 1. Software Developer | 370,600 | 120,730 | Bachelor's |
| 2. Nurse Practitioner | 112,700 | 120,680 | Master's |
| 3. Medical and Health Services Manager | 136,200 | 101,240 | Bachelor's |
| 4. Physician Assistant | 38,400 | 121,530 | Master's |
| 5. Information Security Analyst | 56,500 | 102,600 | Bachelor's |
| 6. Physical Therapist | 40,400 | 95,620 | Doctorate |
| 7. Financial Manager | 123,100 | 131,710 | Bachelor's |
| 8. IT Manager | 82,400 | 159,010 | Bachelor's |
| 9. Web Developer | 28,900 | 77,030 | Bachelor's |
| 10. Dentist | 7,700 | 160,370 | Doctorate |
| 11. Lawyer | 80,200 | 127,990 | Doctorate |
| 12. Veterinarian | 16,800 | 100,270 | Doctorate |
| **Summary: Medical related jobs** | **352200** | | **Master/Doctor** |
| **Summary: Computing related jobs** | **538400** | **x 1.53** | **Bachelor** |

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# 25 Best Jobs US News 2024

https://money.usnews.com/money/careers/articles/the-25-best-jobs

| Job Title | Median Salary | New Job # | Job growth % |
|---|---|---|---|
| 1. Nurse Practitioner | $121,610 | 118,600 | 45.5% |
| 2. Financial Manager | $139,790 | 126,600 | 16% |
| 3. Software Developer | $127,260 | 410,400 | 25.7% |
| 4. IT Manager | $164,070 | 85,900 | 15.4% |
| 5. Physician Assistant | $126,010 | 39,300 | 26.6% |
| 6. Medical and Health Services Manager | $104,830 | 144,700 | 28.4% |
| 7. Information Security Analyst | $112,000 | 53,300 | 31.6% |
| 8. Data Scientist | $103,500 | 59,300 | 35.1% |
| 9. Actuary | $113,990 | 6,900 | 23% |
| 10. Speech-Language Pathologist | $84,140 | 33,100 | 19.3% |
| 11. Marketing Manager | $140,040 | 23,700 | 6.6% |
| 12. Statistician | $98,920 | 10,600 | 31.8% |
| **Summary: Medical related jobs** | **$105,074** | **335,700** | |
| **Summary: Computing related jobs** | **$121,150** | **619,500** | **x 1.85** |

# 25 Best Jobs US News 2025

https://money.usnews.com/money/careers/articles/the-25-best-jobs

| Job Title | Median Salary | New Job # | Unemploy rate |
|---|---|---|---|
| 1. Nurse Practitioner | $126,260 | 135,500 | 0.6% |
| 2. IT Manager | $169,510 | 106,900 | 1.4% |
| 3. Physician Assistant | $130,020 | 43,700 | 1.6% |
| 4. Financial Manager | $156,100 | 138,300 | 1.4% |
| 5. Software Developer | $132,270 | 303,700 | 2.4% |
| 6. Information Security Analyst | $120,360 | 59,100 | 0.5% |
| 7. Medical and Health Services Manager | $110,680 | 160,600 | 1.5% |
| 8. Data Scientist | $108,020 | 73,100 | N/A |
| 9. Speech-Language Pathologist | $89,290 | 33,300 | 1.6% |
| 10. Actuary | $120,000 | 6,600 | N/A |
| 11. Marketing Manager | $157,620 | 31,700 | 3.3% |
| 12. Veterinarian | $119,100 | 16,800 | 0.2% |
| **Summary: Medical related jobs** | **$121,515** | **356,600** | |
| **Summary: Computing related jobs** | **$108,432** | **549,400** | **x 1.54** |

Why do you want to be a software architect/ engineer?

How do you become a software architect/ engineer, instead of a programmer?

# Lecture 2-2 Outline



## How do you become a Software Architect/Engineer?

Enterprise Architecture and Business Process

- Business and Technology Interaction
- Enterprise Architecture Integration (EAI)
- Enterprise Architecture Framework (EAF)
- Business Process Management (BPM)
- Development Life Cycle
- Dynamic Application Architecture for Dynamic Business

# eBusiness and Technology Interaction

**e-business**

| Computerized establishment of e-business with basic information exchange with clients | Maintain a multi page Web site; Use Web-form and Web-email for data exchange with clients | Core business on Web with database and Web-enabled execution of business process and transaction. | Enterprise level collaboration & integration, Dynamic business composition, EAI, BPM | User-provided contents, Dynamic discovery, Social networking, Social integration | *On Demand e-business with AI-based real-time reconfigure-ation, resilient, & automated evolution* |
|---|---|---|---|---|---|

**Technology**

| Early stage of Internet: Email FTP | Static Web 1.0 with HTML forms and web mail support | Web-based computing model and Web database integration | Web 2.0, WSDL, Web 3.0, SOA, SOC, SOD, ESB, Dynamic application architecture | Web 4.0*, HTML5, REST services, Cloud computing, Big Data, Social computing and networking platforms | *Web 5.0* with proactive artificial intelligent services and applications* |
|---|---|---|---|---|---|
| 80s | 90s | 00-04 | 05-09 | 10-19 | 20- |

* From technology perspective
From user's perspective: Web 3.0

# Enterprise Architecture Integration (EAI)

Enterprise level collaboration & integration, Dynamic business composition, EAI, BPM

- Architecture: The fundamental organization of a system, its components, and the relationships among the components;
- Enterprise Architecture: An architecture of a system involves one or more enterprises, including their business processes, data and information, as well as the technologies;
- Enterprise Architecture Integration (EAI): Seamless integration of architectures, business processes, data and information, as well as the infrastructure / technologies within an enterprise and across enterprises.

# Purposes of EAI

Google EAI jobs

- Architecture Integration: Defining architecture framework for managing the development and maintenance of architecture descriptions.
- Data (information) Integrity: Ensuring that information in multiple systems is kept consistent.
- Application Integration: Different applications can communicate with each other, and are ready for integration.
- Vendor Independence: If a business application is replaced with a different vendor's application, the business rules do not have to be re-implemented.
- Common Façade: An EAI system could be a cluster of different applications. It can provide a single consistent access interface to these applications and shielding users from having to learn to interact with different software packages.

# Enterprise Architecture Integration (EAI)

- EAI is composed of a collection of technologies and services.
- A middleware to enable integration of systems and applications within and across the enterprises.
- Examples
  - ASU Website, involving application, admission, students database, course delivery, financial management, …;
  - Supply chain management applications (for managing inventory; shipping, and interaction with other enterprises);
  - Customer relationship management applications (for managing current and potential customers);
  - Business intelligence applications (for finding patterns from existing data generated from operations). Big data and machine learning are major tools for today's business intelligence applications.

# Enterprise Architecture Framework (EAF)

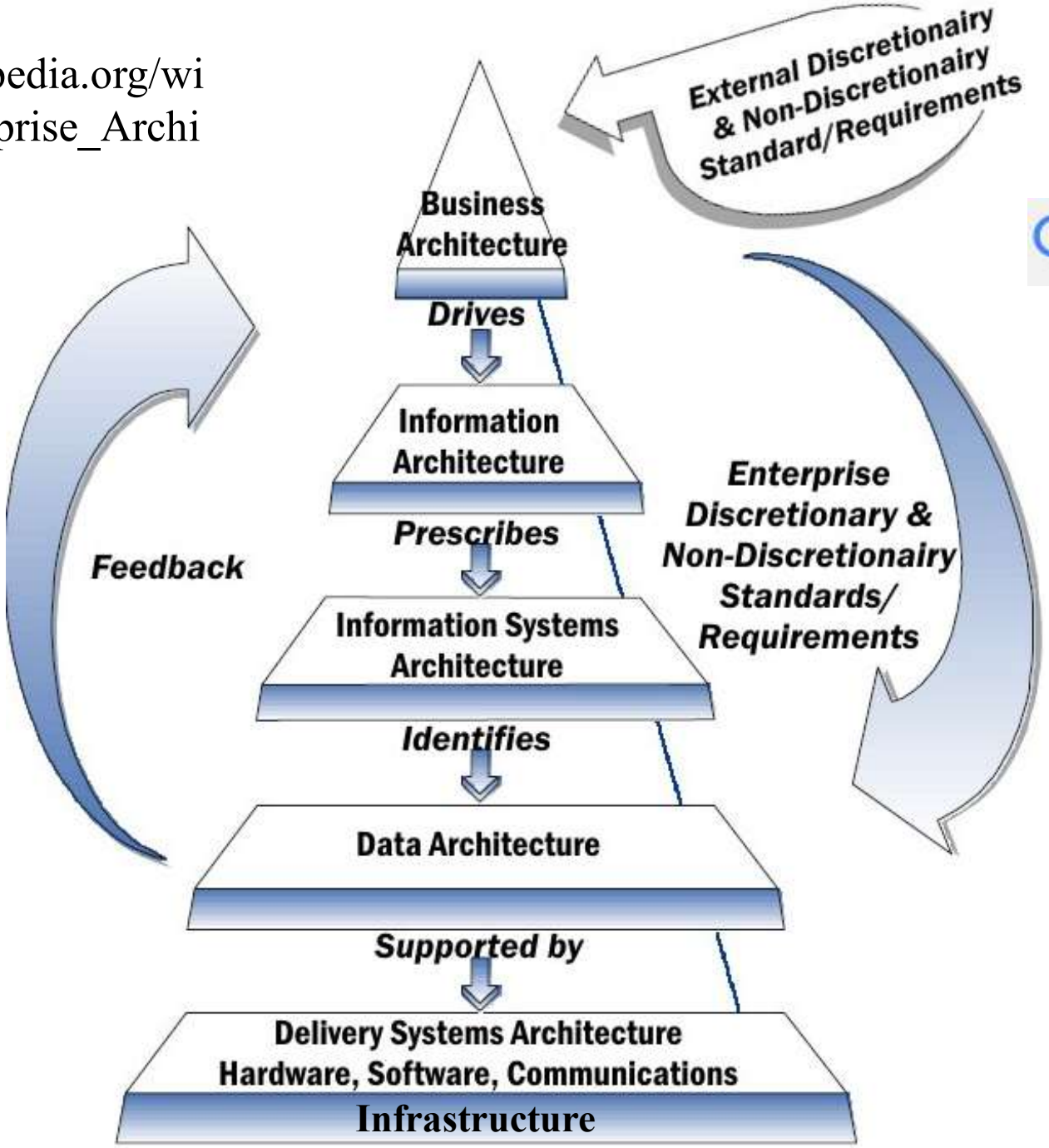
Google EAF jobs

- EAF is a standard developed by the U.S. NIST, dealing with architecture integration.
- EAF is an organizing mechanism for managing the development and maintenance of architecture descriptions.
- EAF provides a structure for organizing resources and describing related activities.
- EAF follows Enterprise Architecture Model (EAM)
- The other layers of integration in EAM (data, application, and infrastructure) will be discussed in the following lectures.

# National Institute of Standards and Technology (NIST)

# Enterprise Architecture Model (EAM)

https://en.wikipedia.org/wiki/NIST_Enterprise_Architecture_Model

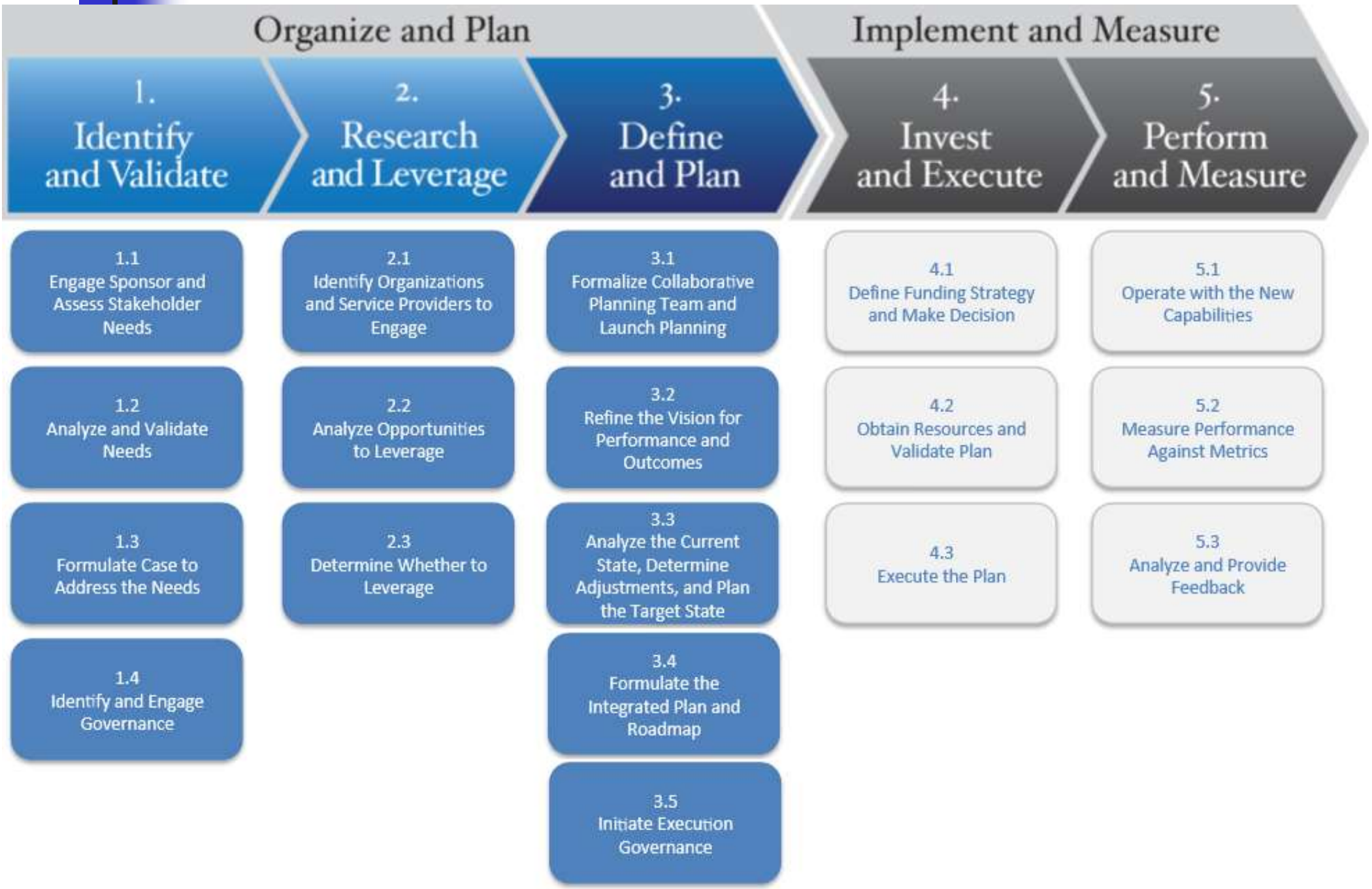


# Federal Enterprise Architecture Framework (FEAF)

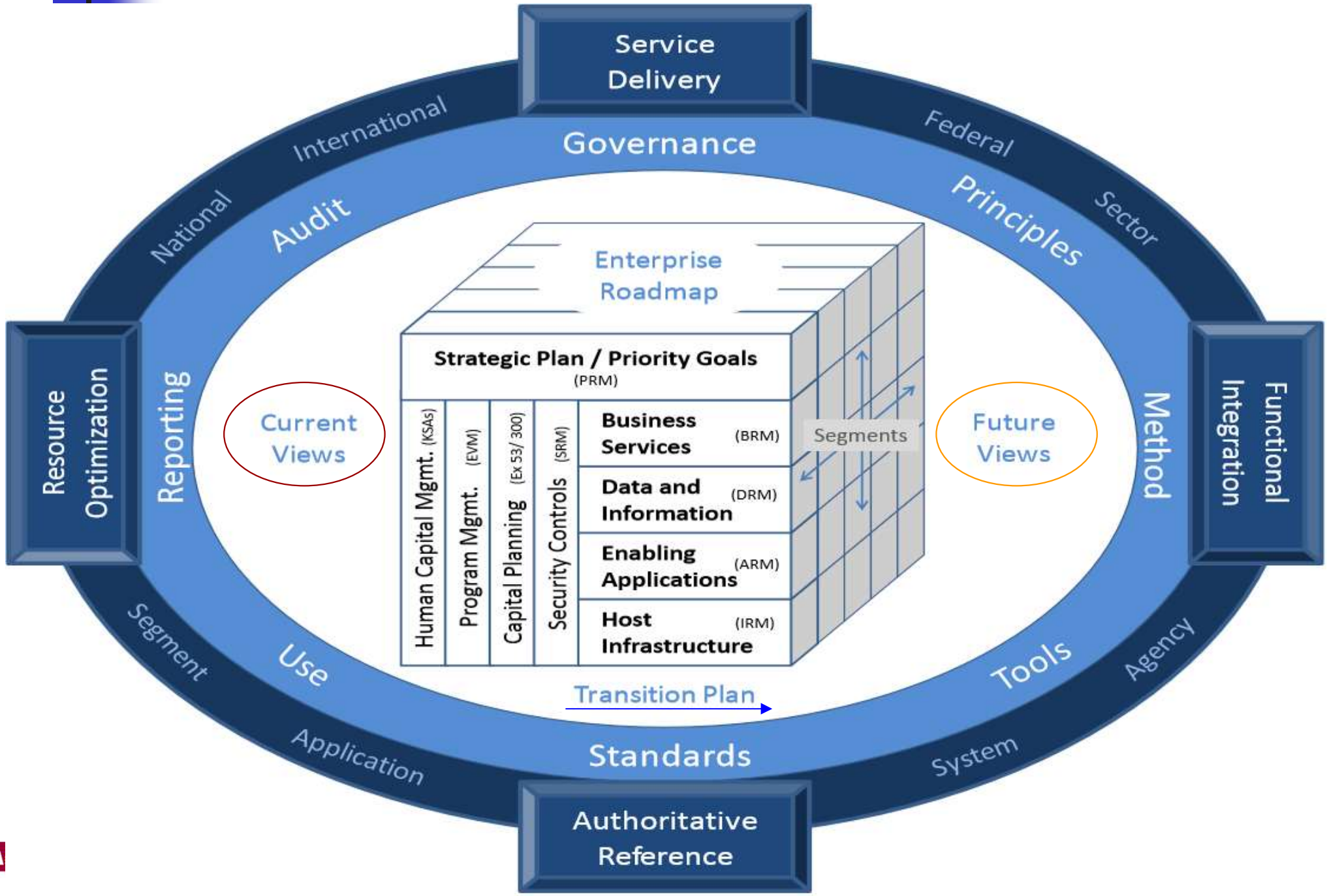
https://obamawhitehouse.archives.gov/sites/default/files/omb/assets/egov_docs/fea_v2.pdf 2013, 434 pages

- Developed by The Chief Information Officers Council
- Follow the NIST Enterprise Architecture Model
- Purposes of developing this framework:
  - Organize Federal information and promote Federal interoperability;
  - Promote information sharing among Federal organizations;
  - Help Federal organizations developing their architectures;
  - Help Federal organizations quickly developing their IT investment processes;
  - Serve customer needs better, faster, and more cost effectively;
  - Provide potential for Federal and Agency to reduce costs.

# Federal Enterprise Architecture Framework (FEAF) Development in Five Steps

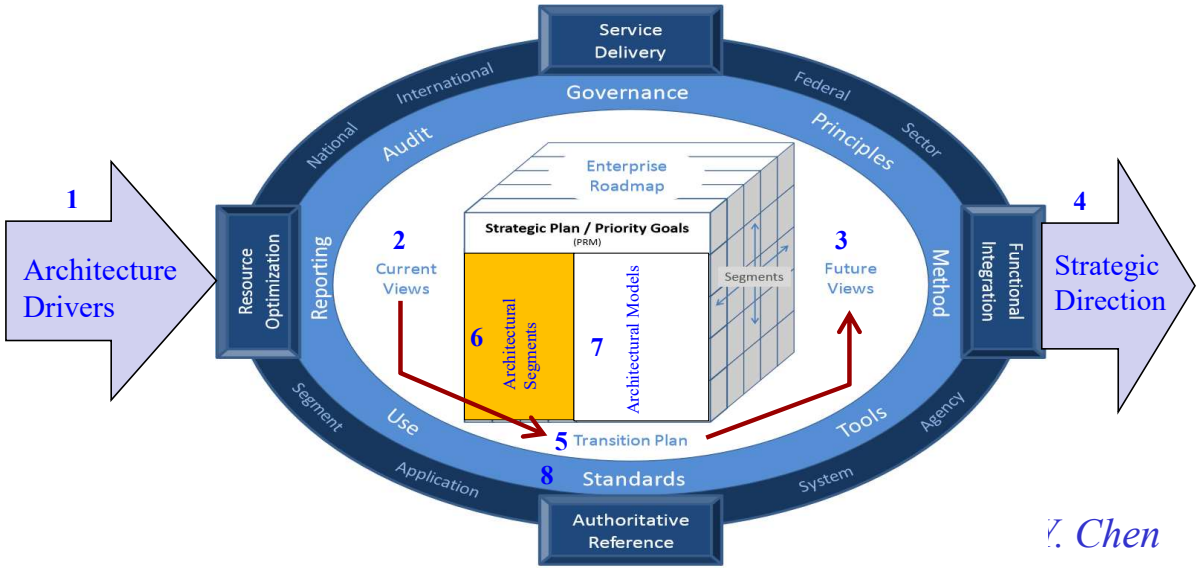


# Federal Enterprise Architecture Framework (FEAF) Organization



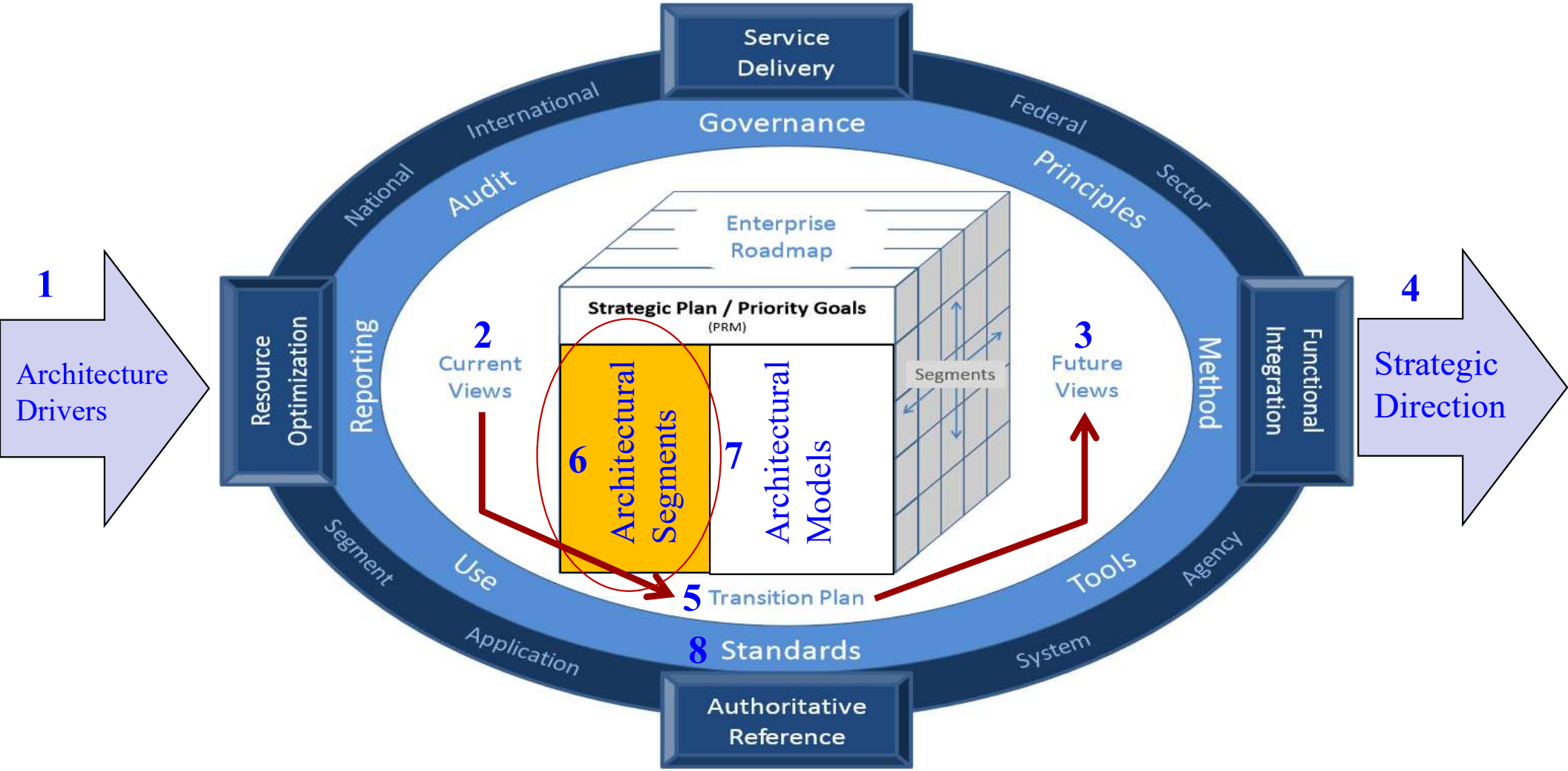
# Eight Key Components of the FEAF

1. Architecture Drivers: business and design stimuli
2. Current Architecture (View): the "as is" enterprise architecture
3. Future Architecture (View): the "to-be-built" enterprise architecture
4. Strategic Direction – Overall guidelines to the development from current architecture to the target architecture
5. Transition Plan – Concrete support to the migration from the current to the target architecture
6. Architectural Segments - Enterprises within the total Federal Enterprise.
7. Architectural Models - business and design models that describe the segments of the enterprise
8. Standards - All standards and references (some of which may be mandatory), and best practices



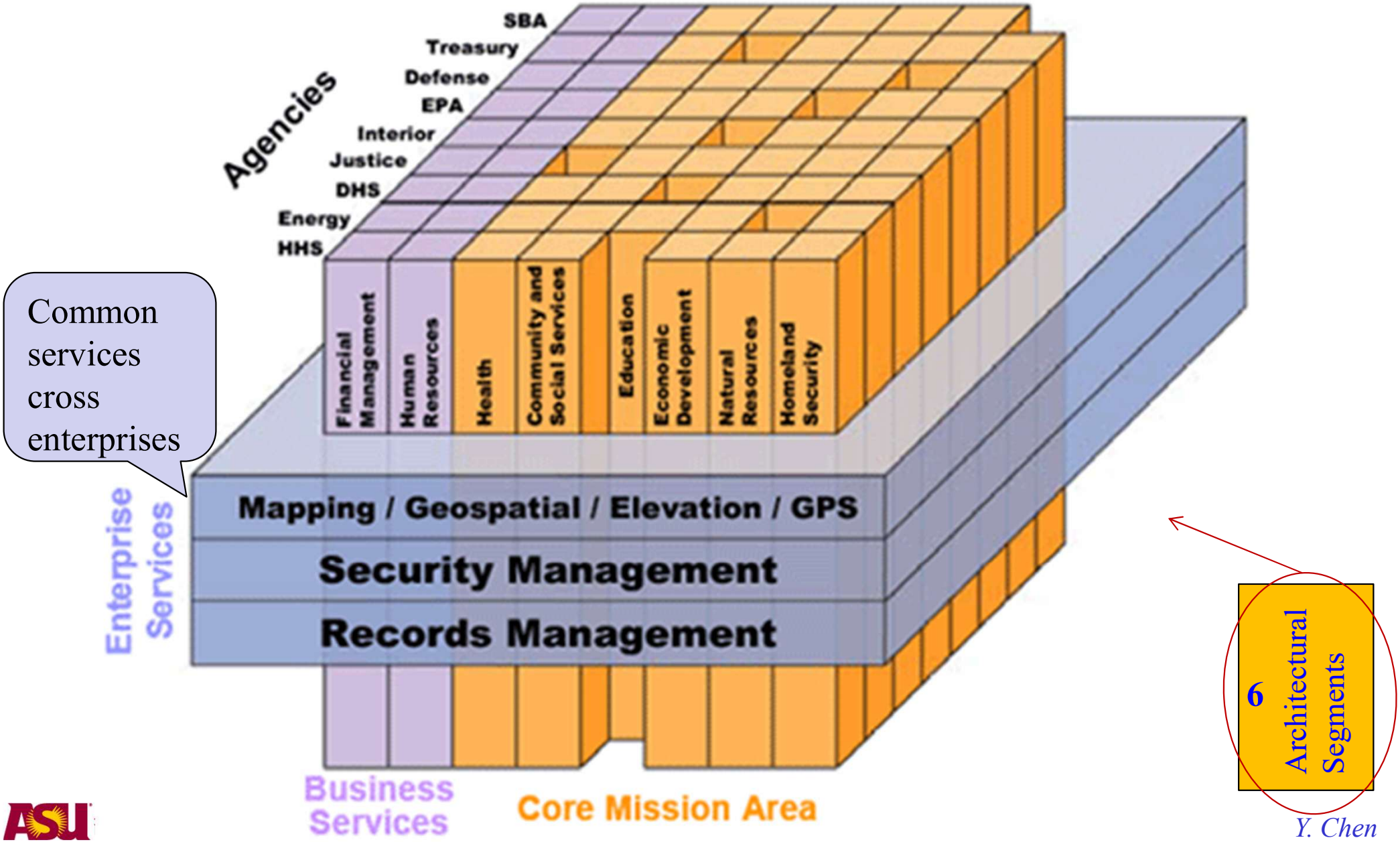
# FEAF Level I View (from 20,000 feet)

- The eight components needed for developing and maintaining the Federal Enterprise Architecture



# Architectural Segments in FEAF

Each architectural segment is an enterprise within the total Federal Enterprise.

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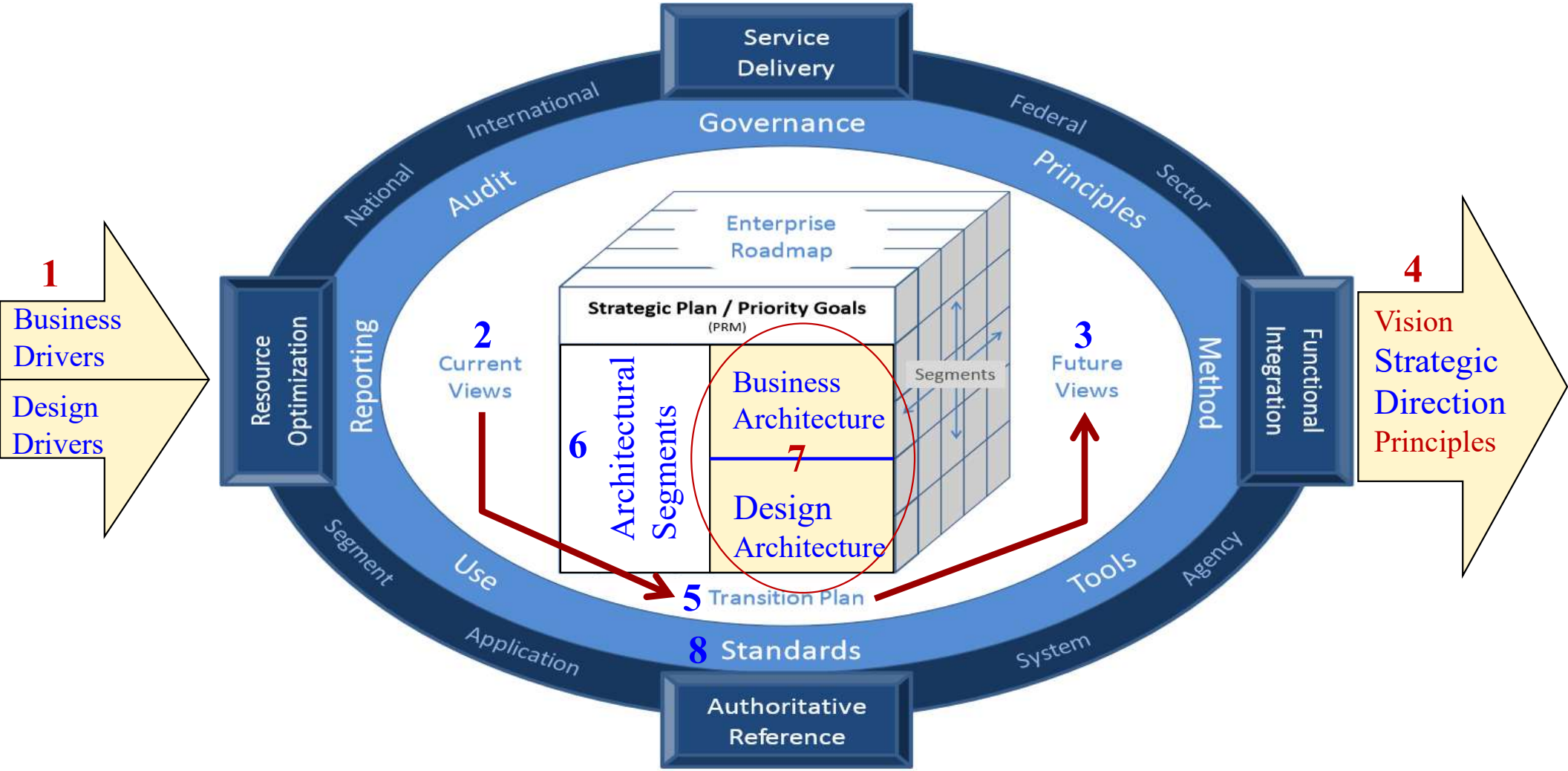
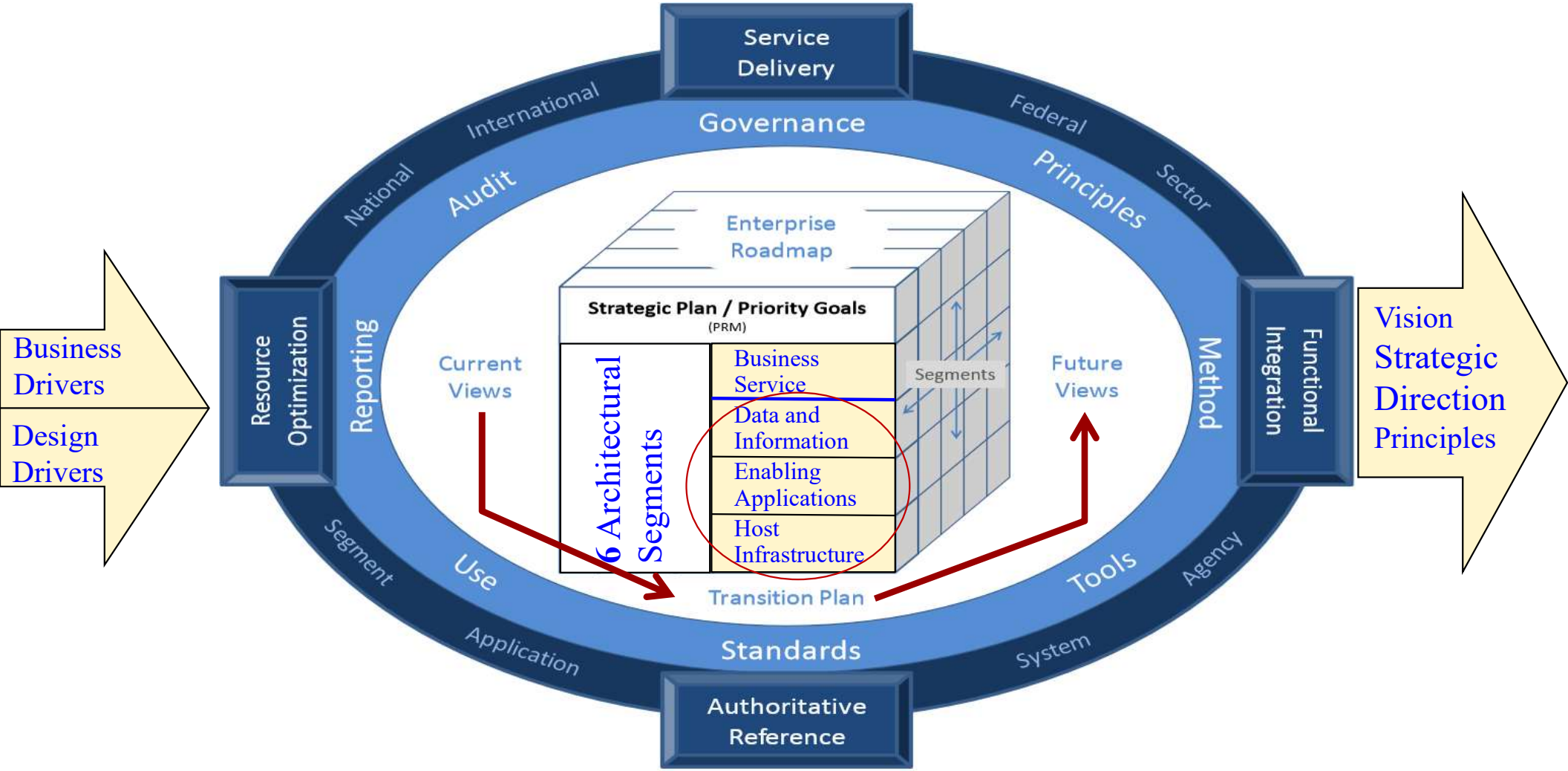
# FEAF Level II View (from 10,000 feet)

- The business and design pieces of the Federal Enterprise Architecture and how they are related



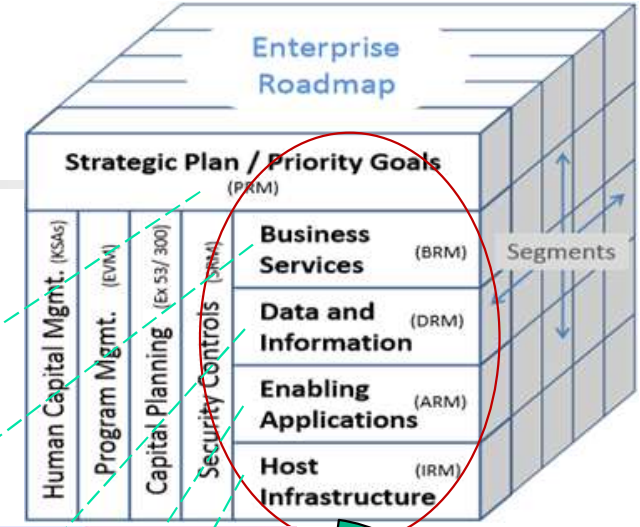
# FEAF Level III View (from 5,000 feet)

- Expand design pieces of the framework to show the three design architectures: data, applications, & infrastructure



# Architecture Standards and Reference Models

FEAF design is based on Architecture Standards and Reference Models




**Performance Reference Model** (PRM)
- Cross-Agency and Intra-Agency Goals and Objectives
- Uniquely tailored performance indicators

❖ Goals
❖ Meas. Area
❖ Meas. Category

**Business Reference Model** (BRM)
- Intra- and inter-agency shared services
- Agencies, customers, partners, providers

❖ Mission Sector
❖ Business Function
❖ Service

**Data Reference Model** (DRM)
- Business-focused data standardization
- Cross-agency information exchanges

❖ Domain
❖ Subject
❖ Topic

**Application Reference Model** (ARM)
- Software providing functionality
- Enterprise service bus

❖ System
❖ Application Component
❖ Interface

**Infrastructure Reference Model** (IRM)
- Hardware providing functionality
- Hosting, data centers, cloud, virtualization

❖ Platform
❖ Network
❖ Facility

**Security Reference Model** (SRM)
- Risk-adjusted security /privacy protection
- Security control design /implementation

❖ Purpose
❖ Risk
❖ Control

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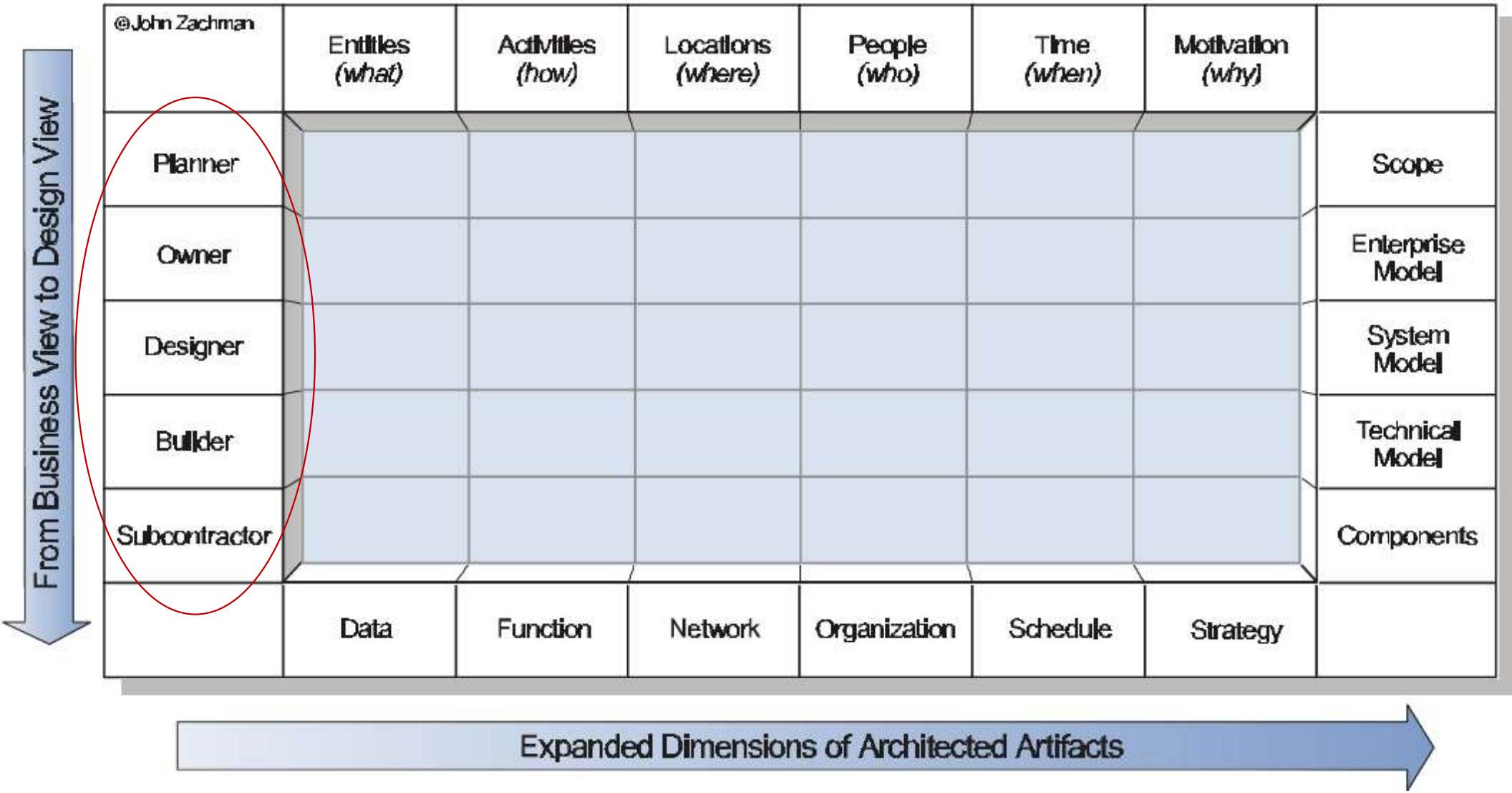
# FEAF Level IV View (from 500 feet)

- How is the business architecture supported by the three design architectures based on architecture reference models?
  - Zachman，EAP，TOGAF
- Zachman Model: *IBM Systems Journal* 1987 article: "A Framework for Information Systems Architecture":
  - It is a model for bridging the business architecture to the three design models (data, application, and infrastructure models).
  - The cost involved and the success of the business depend increasingly on its information systems, and a disciplined approach to the management of those systems is required.
  - Architecture framework is simply a logical structure for classifying and organizing the descriptive representations of an enterprise that are significant to the management of the enterprise, as well as to the development of the enterprise's systems.

# FEAF Level IV View Based on Zachman Model

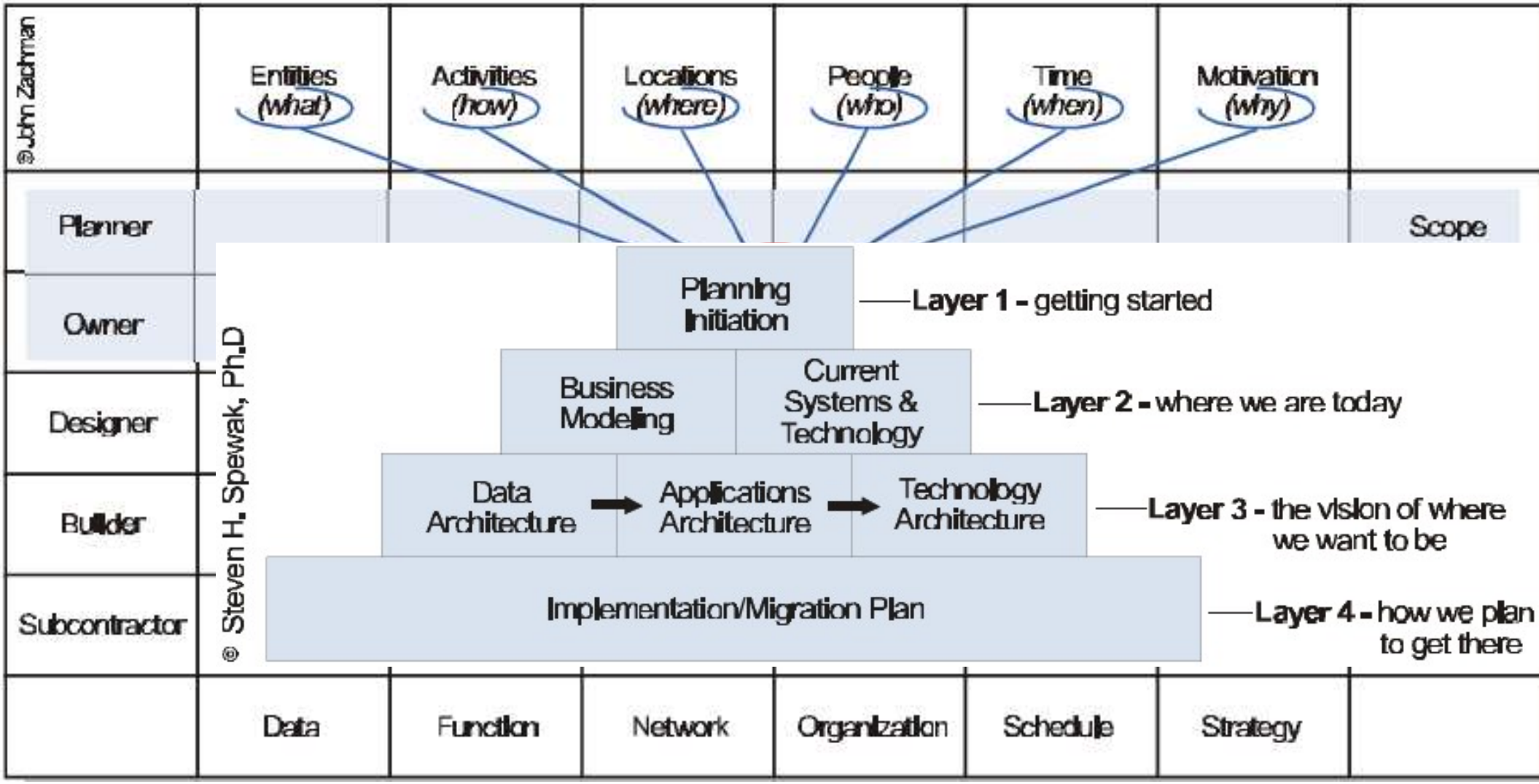
- The Framework enables communication among the various participants involved in developing or changing the structure



# From Zachman Model to EAP

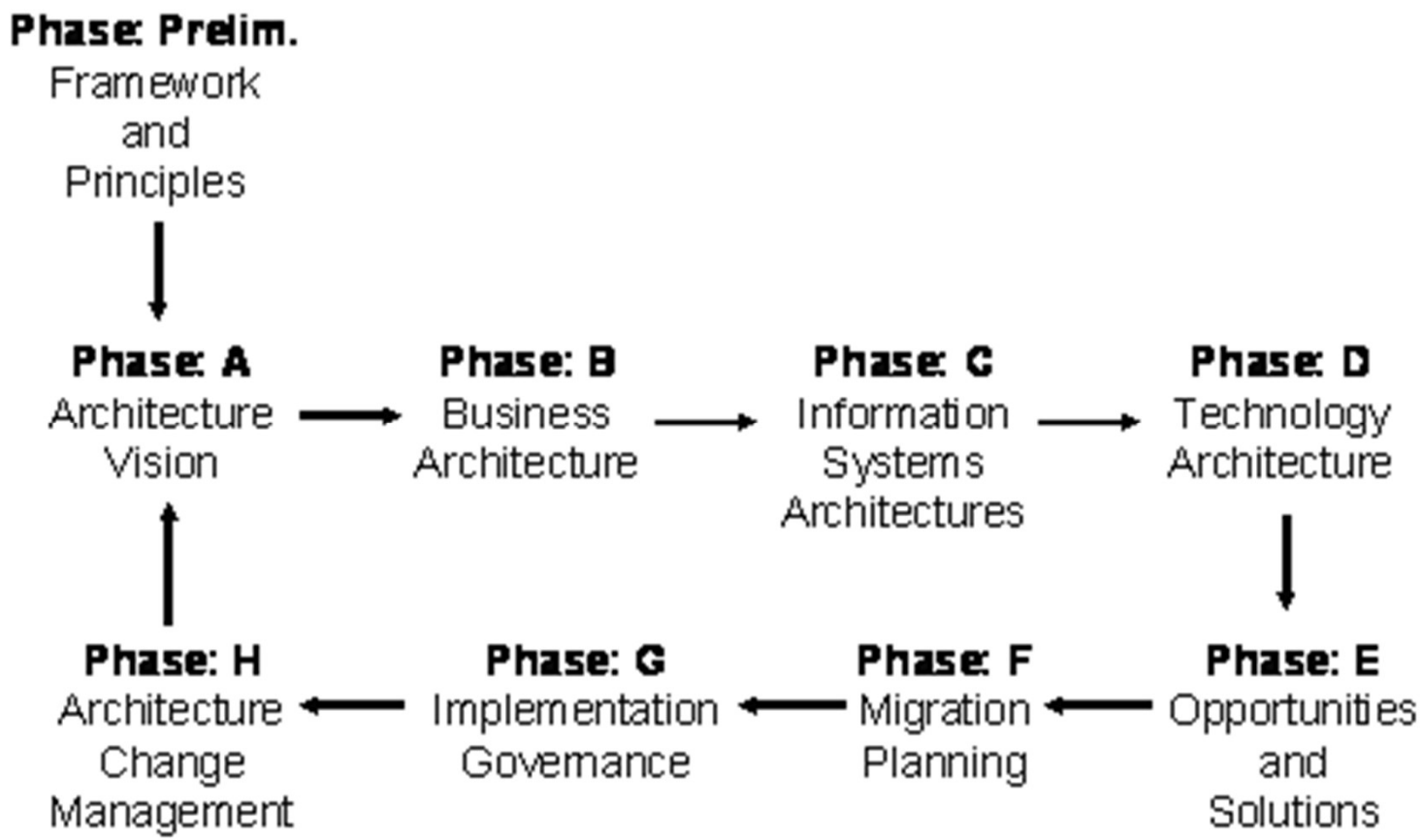
- The Zachman Model tells what need to be done, how, where, when, why, and by whom.
- It categorizes and organizes the artifacts: data, function, network, organization, schedule, and strategy.
- Enterprise Architecture Planning (EAP), proposed by Dr. Steven Spewak,
  - defines the process of completing Zachman model;
  - focuses on defining what data, applications, and technology architectures are appropriate for supporting the overall enterprise.

# Enterprise Architecture Planning (EAP)



# Open Group Architecture Framework (TOGAF)

- TOGAF describes another architectural process;
- It elaborates Zachman model. Zachman tells you how to categorize your artifacts. TOGAF gives you a process for creating them;



# BPM: Business Process Management

http://en.wikipedia.org/wiki/Business_process_management

- is an EAI technology at the design/process level;
- is a management approach focusing on aligning all aspects of an organization with the wants and needs of clients;
- allows organizations to abstract business process from technology infrastructure;
- goes beyond automating business processes or solving business problems using software;
- enables business to respond to changing consumer, market, and regulatory demands faster than competitors - creating a competitive advantage.

# BPM Lifecycle (Before Service Orientation)

- **Process Design:** encompasses both the identification of existing processes and the design of "to-be" processes.
- **Modeling:** takes the theoretical design and introduces combinations (relationships) of variables or parameters;
- **Execution:** develops an application that executes the required steps of the process;
- **Monitoring:** tracks processes and statistics on the performance of one or more processes;
- **Optimization:** retrieves information from modeling or monitoring phase; identifying the potential or actual bottlenecks and the potential opportunities for cost savings or other improvements; and then, applying those enhancements in the design of the process.

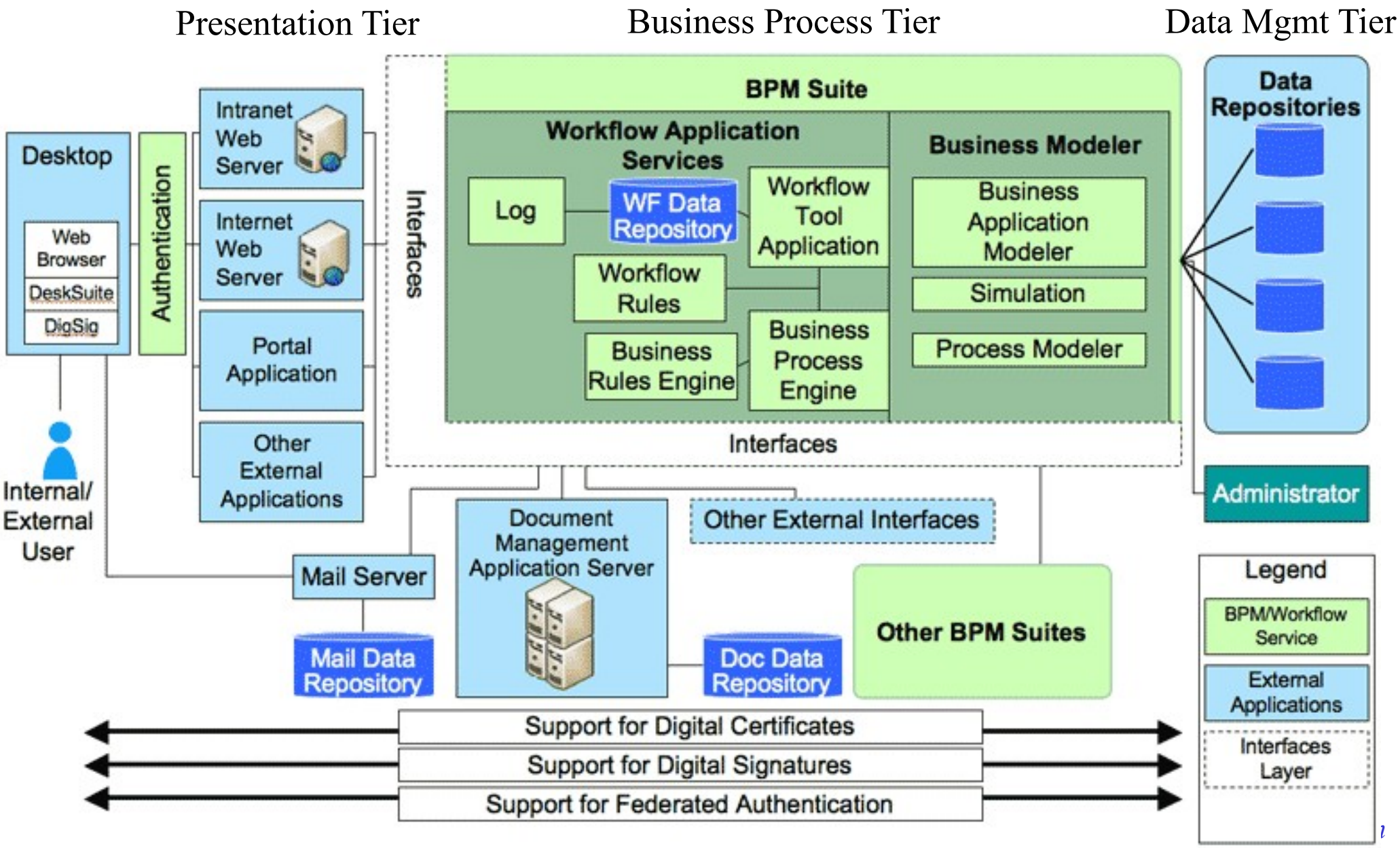
# BPM Suite (BPMS) Development Framework

BPM Suite contains four critical components

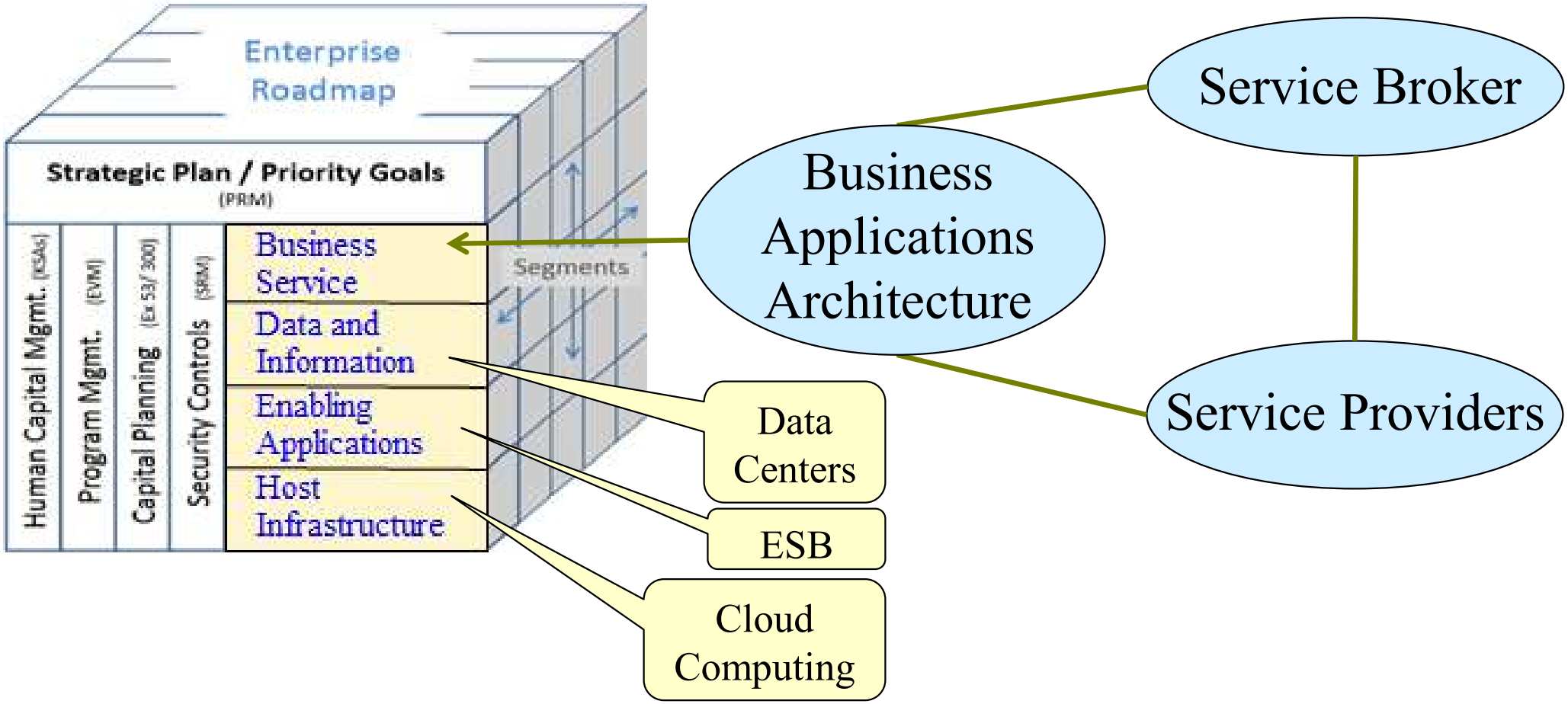
- Process Engine – a robust platform for modeling & executing process-based applications and business rules;
- Business Analytics — enables managers to identify business issues, trends, and opportunities with reports and dashboards and react accordingly;
- Content Management — provides a system for storing & securing documents, images, and other files;
- Collaboration Tools — removes intra- and interdepartmental communication barriers through discussion forums, dynamic workspaces, and message boards.

# BPM Suite and Its Implementing a Business Process

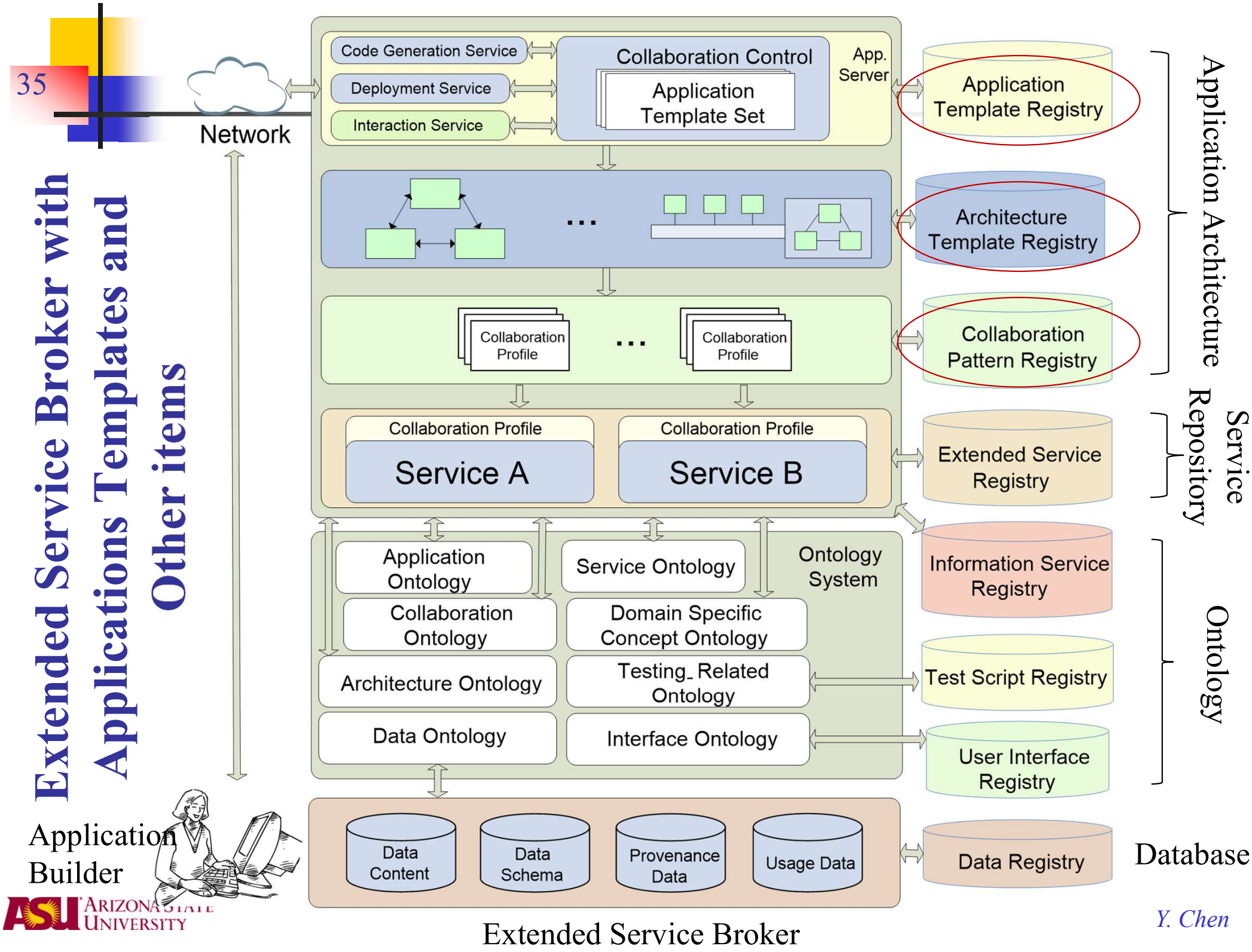
http://en.wikipedia.org/wiki/Business_process_management



# Service-Oriented Applications Architecture (SOAA) at the Process Level



# Extended Service Broker with Applications Templates and Other items

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# Extended Broker

- Publish architecture templates;
- Publish application template/workflow;
- Publish collaboration templates;

Help in application architecture design

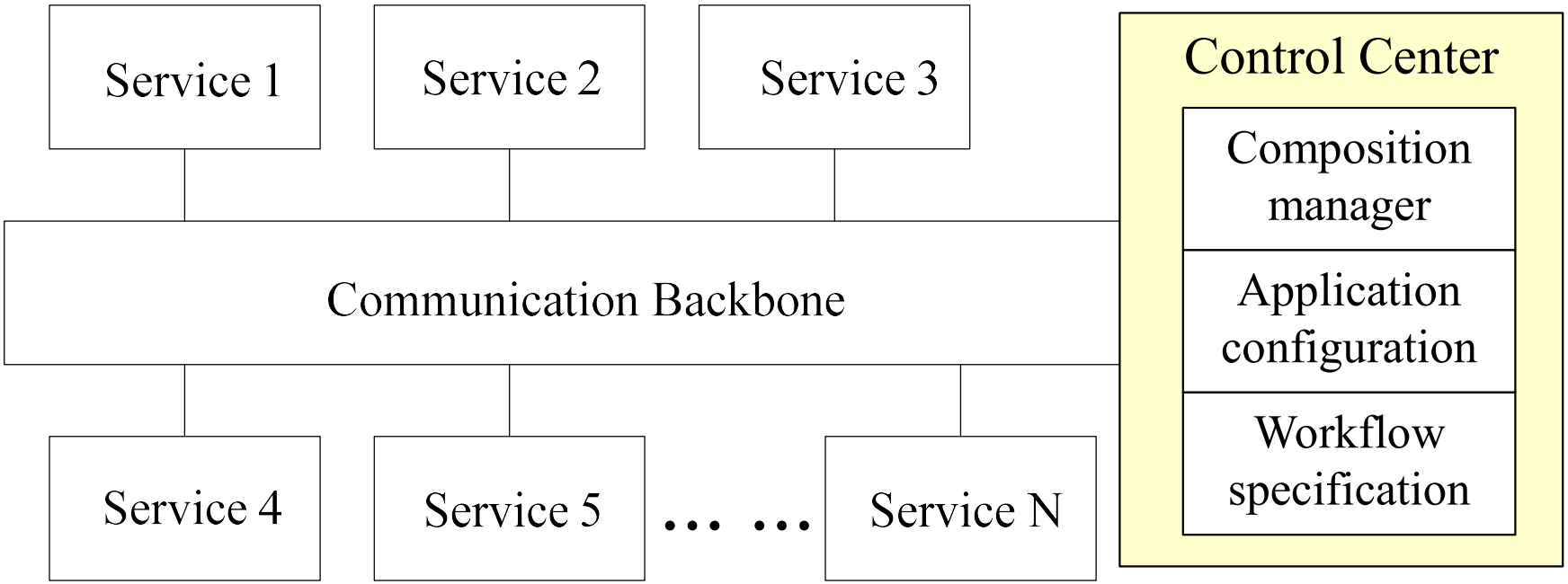
- Publish SOAP services with collaboration templates;
- Publish data and information services (RESTful services)
- Publish test scripts
- Publish user interfaces
- Publish Data: a typical example is Data-Centric SOA (DCSOA) and data center in cloud computing

# Service-Oriented Application Architecture and Case Studies

- Dynamic Reconfigurable Application Architecture Organizing services in application;
- Different architectures:
  - SAP Netweaver
  - Oracle SOA Suite
  - IBM ESB

Web 2.0, Web 3.0, SOA, SOC, SOD, ESB, Dynamic application architecture

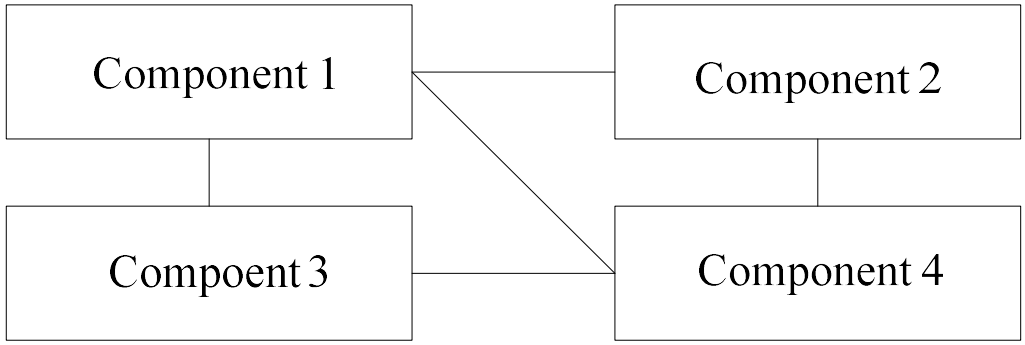
# General Service-Oriented Application Architecture



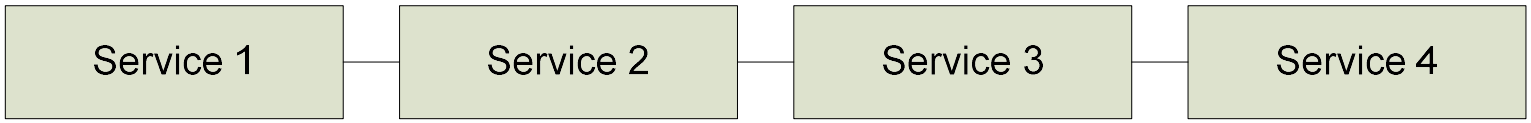

- Composition manager specifies and controls the application composition configuration via a workflow specification.
- The application composition configuration/workflow specification defines how the services shall be connected together to deliver the desired application and how the messages are transferred among the services
- The Communication Backbone is a logic view. It can be a bus or not a bus.

# Static vs. Dynamic Architecture

Static architecture with integrated components




Configuration 1

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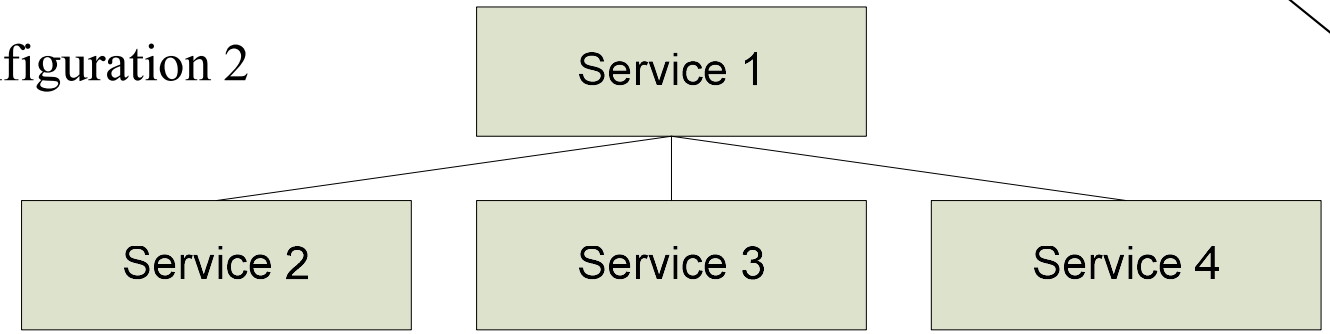
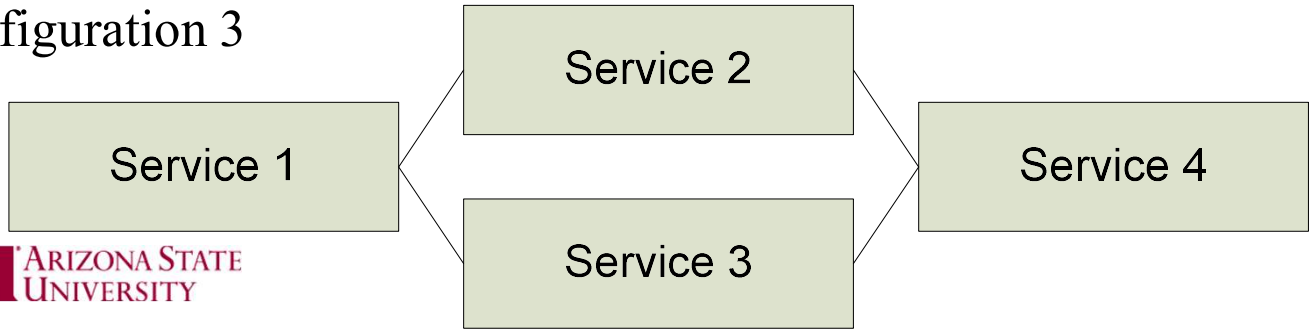
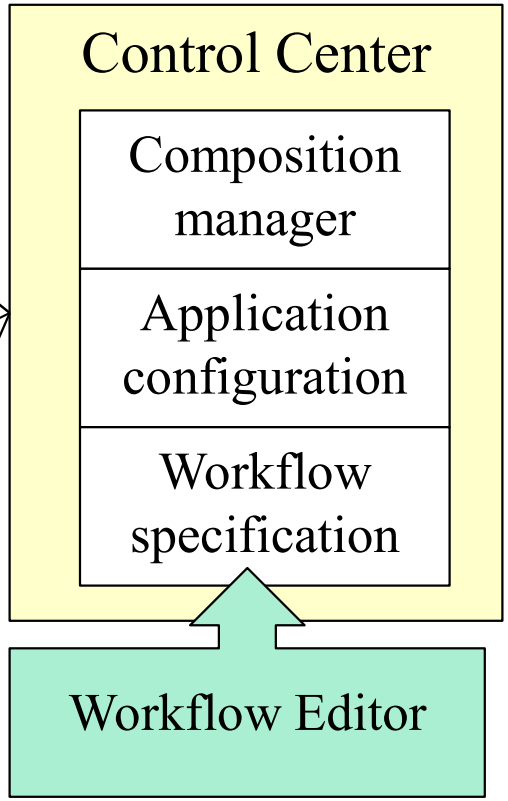
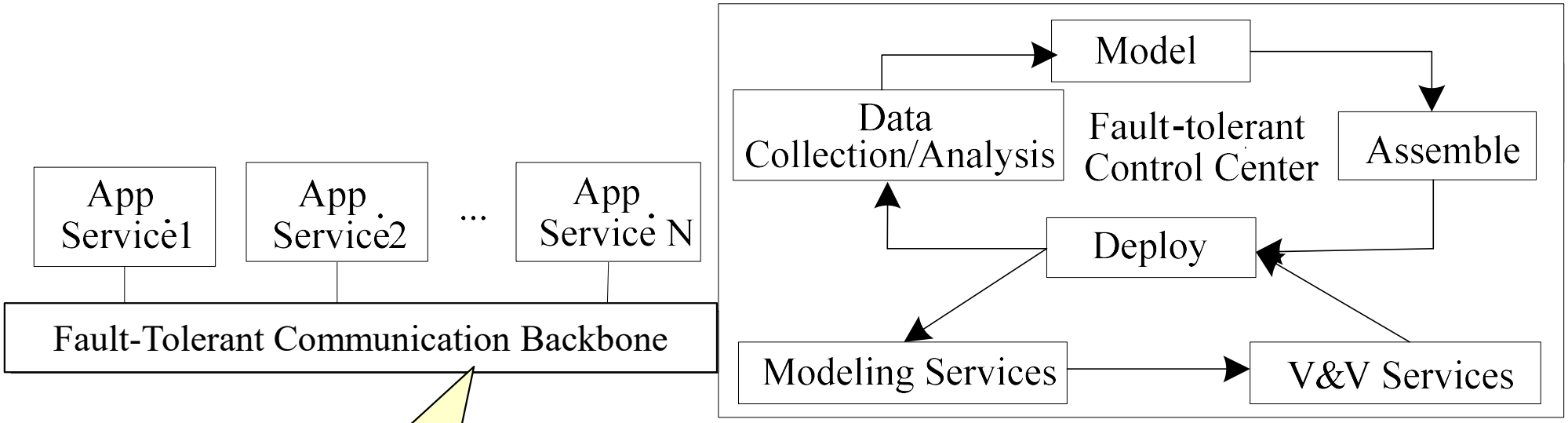



Configuration 3



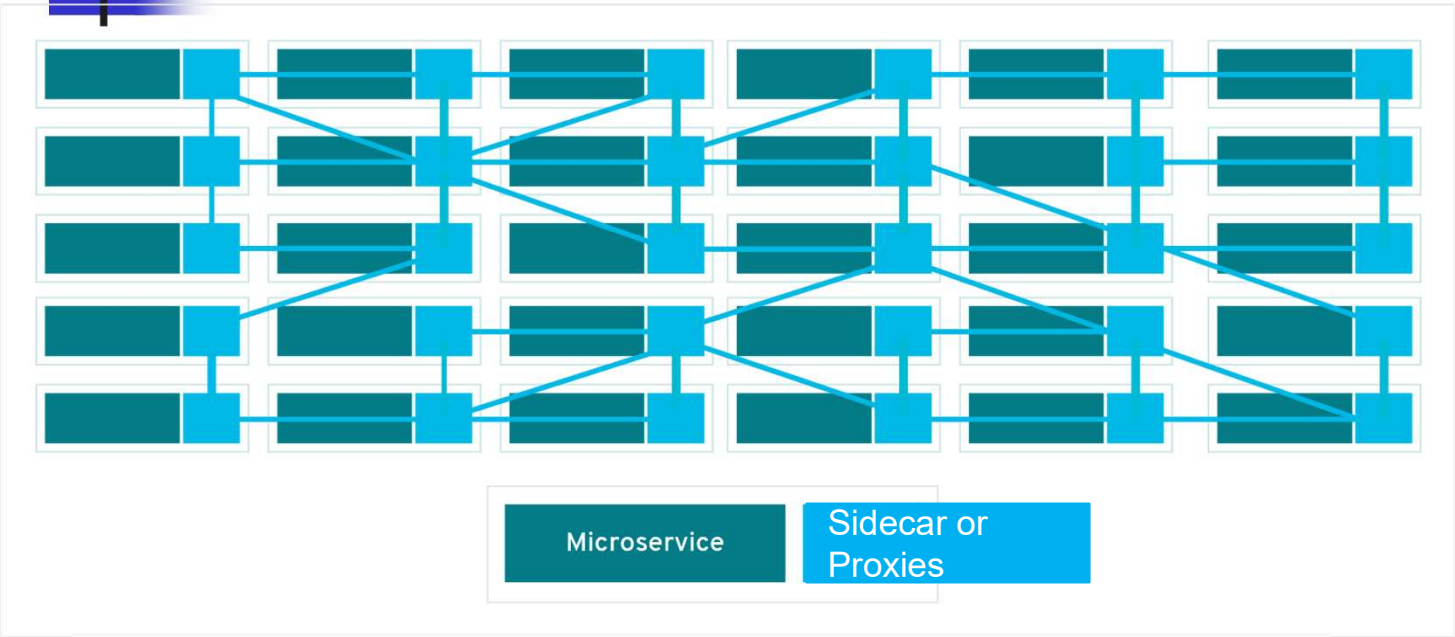



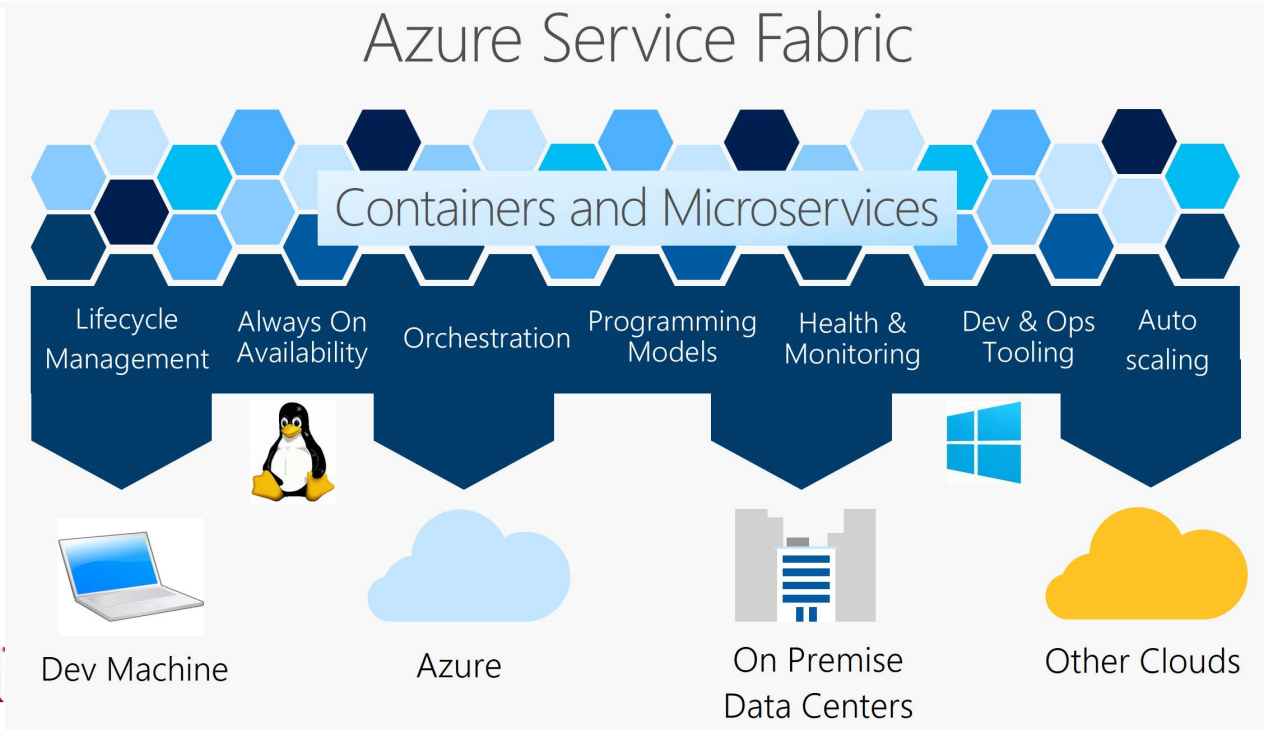
# Lifecycle Reference Model with Dynamic Application Architecture

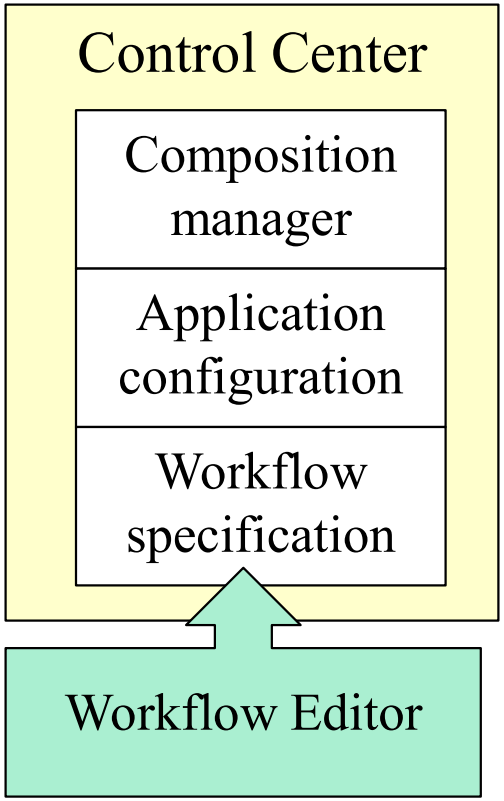



Communication Backbone is a logic view. It can be a bus or not a bus.

# Microservice vs. Dynamic Architecture

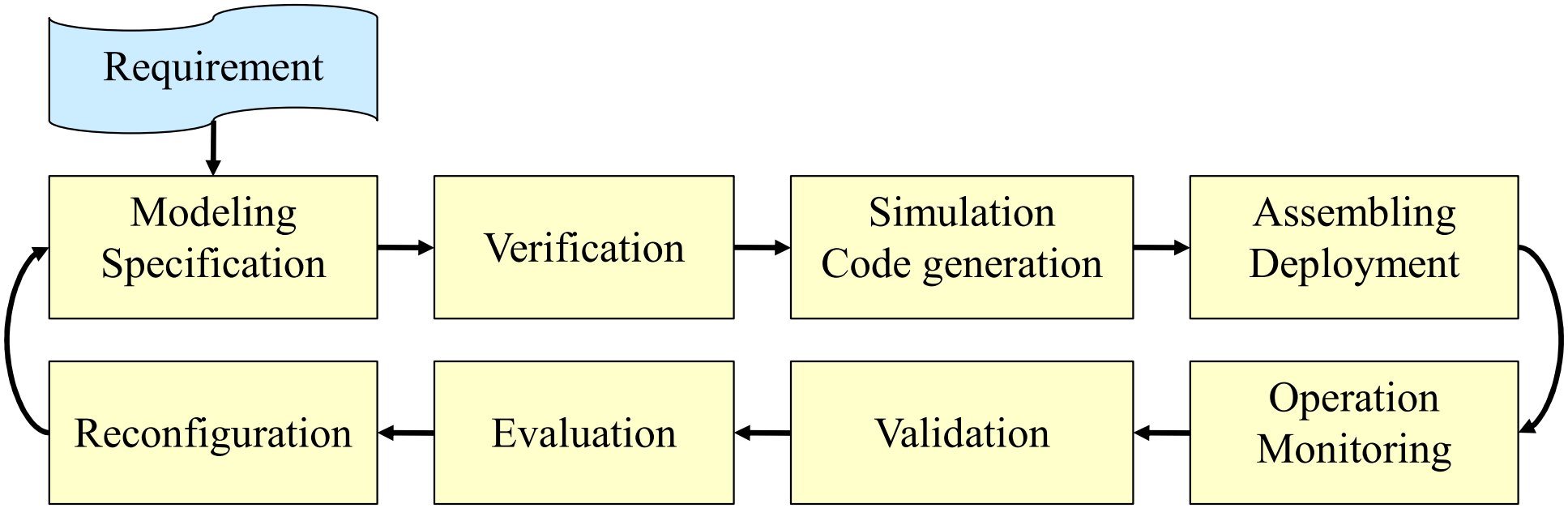


# Lifecycle Reference Model of SOAA

Based on BPM and with additional implementation detail



# Dynamic Architecture Case Studies

- SAP Netweaver
- Oracle SOA Suite
- IBM ESB

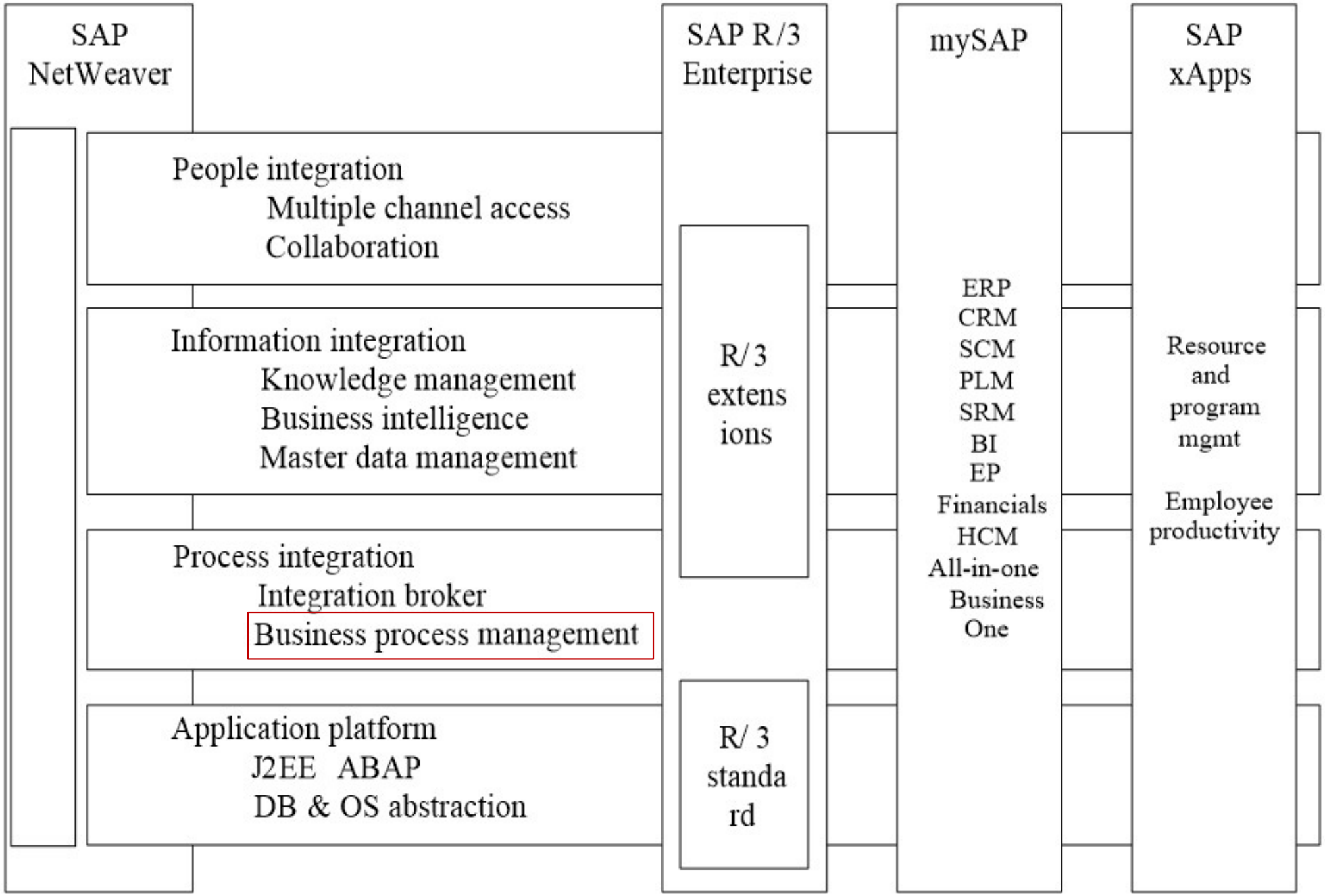
# Application Architecture Development Tools

| Project Name | Architecture Style | Characteristics |
|---|---|---|
| SAP NetWeaver | Layered architecture | Providing integration with the traditional applications, with the SOA applications on Microsoft .Net and IBM WebSphere platforms |
| Oracle Dynamic Business Process | Visual architecture in workflow | Separate architecture from business process; allowing dynamically revising architecture and process on the fly. |
| IBM WebSphere Reference Architecture | Layered Architecture | Merging the SOA application development and the SOA application operation into one single infrastructure based on ESB |
| Microsoft Workflow Foundation | Visual architecture in workflow | Aiming at providing an integrated platform for specifying architecture, modeling, code generation, and execution of SOA applications |

# SAP NetWeaver

- Proposes a concept "Enterprise Services Architecture" (ESA), which provides an architecture framework for process modeling and service applications;
- Provides integration with the traditional applications, such as SAP R/3 and mySAP, with the SOA applications on J2EE, Microsoft .Net, and IBM WebSphere platforms;
- Provides interoperability and integration with other applications by following standards established by W3C.

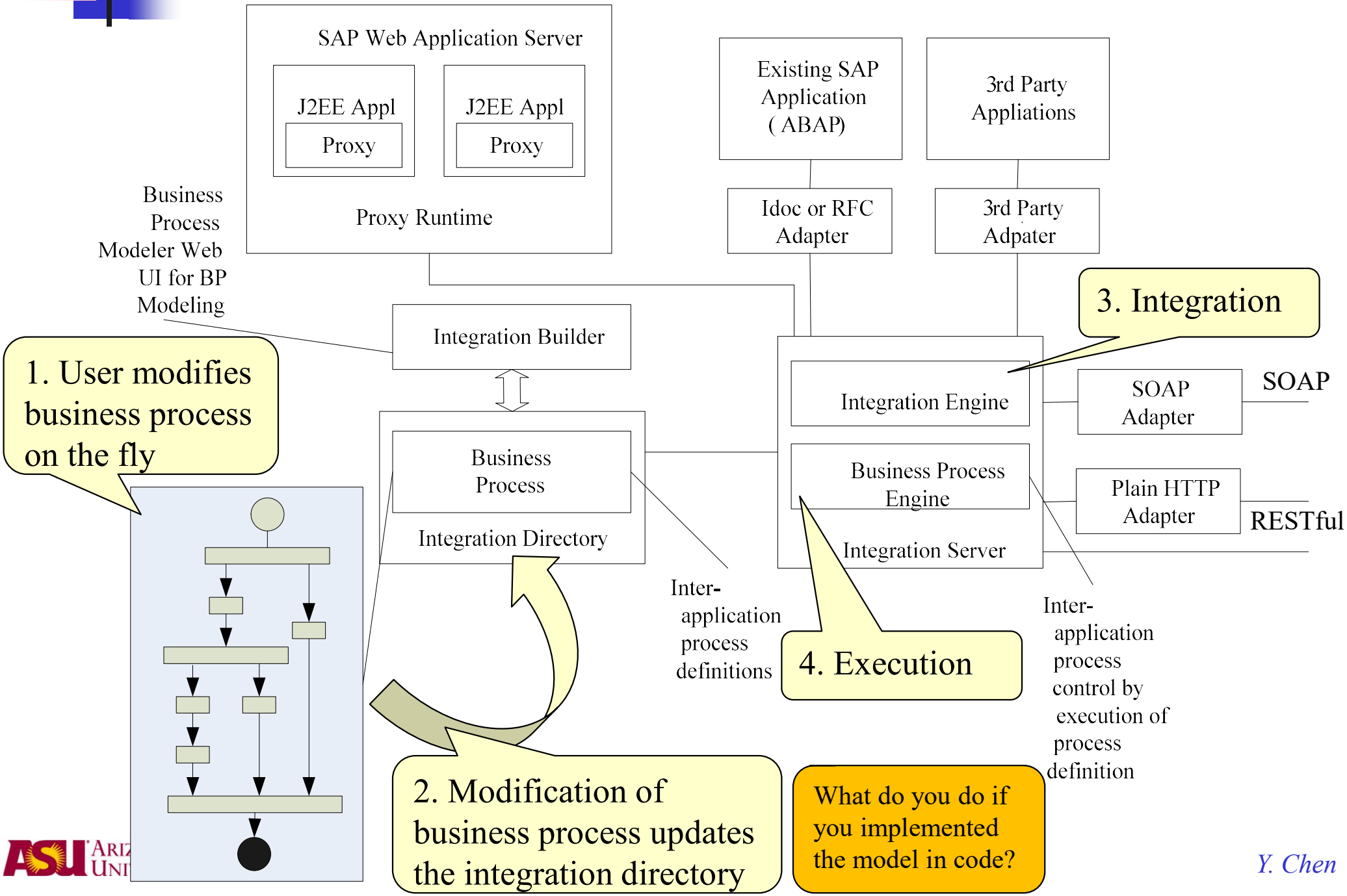
# SAP NetWeaver Architecture



# SAP NetWeaver BPM

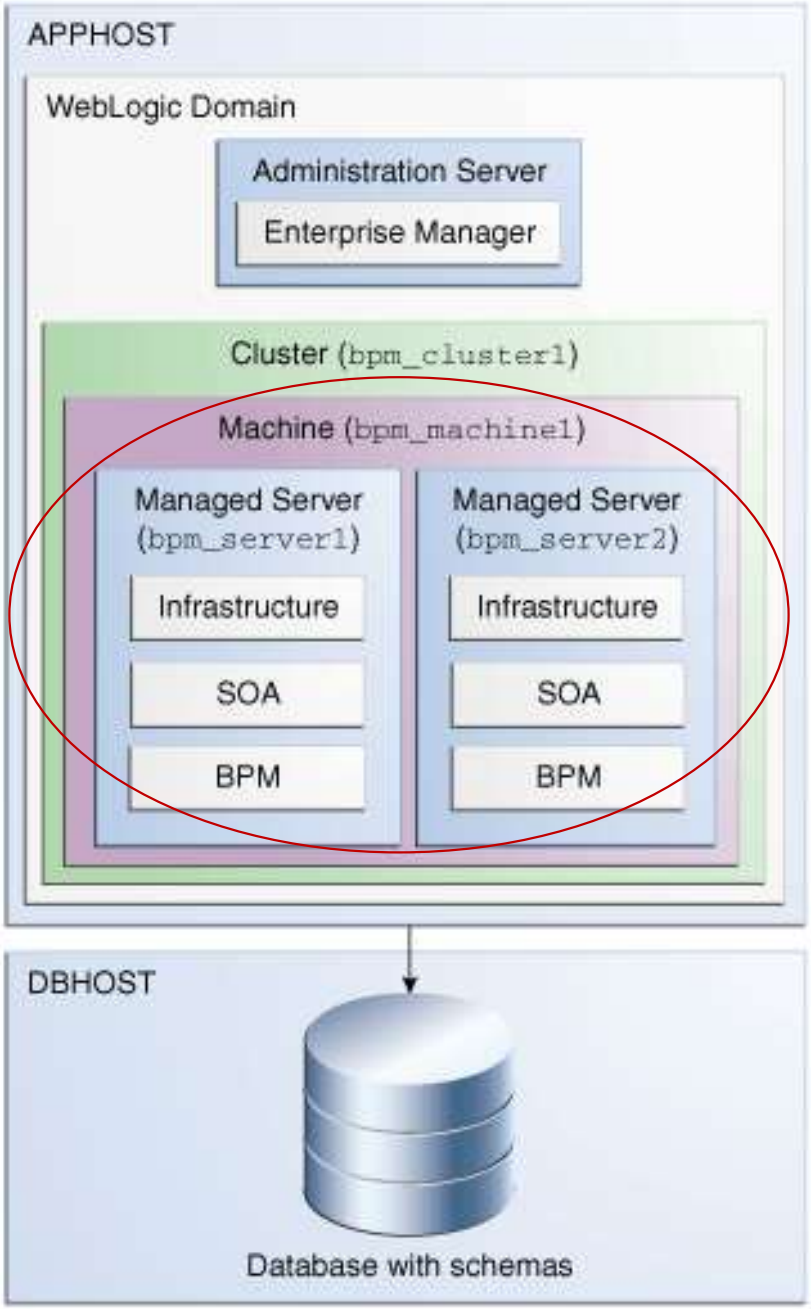
- One of the key modules in SAP Netweaver is the BPM, consisting of the following components:
  - Integration Builder: supports BPEL process
  - Integration Repository,
  - Integration Directory,
  - Business Process Engine

# SAP BPM Supporting Dynamic Application Architecture

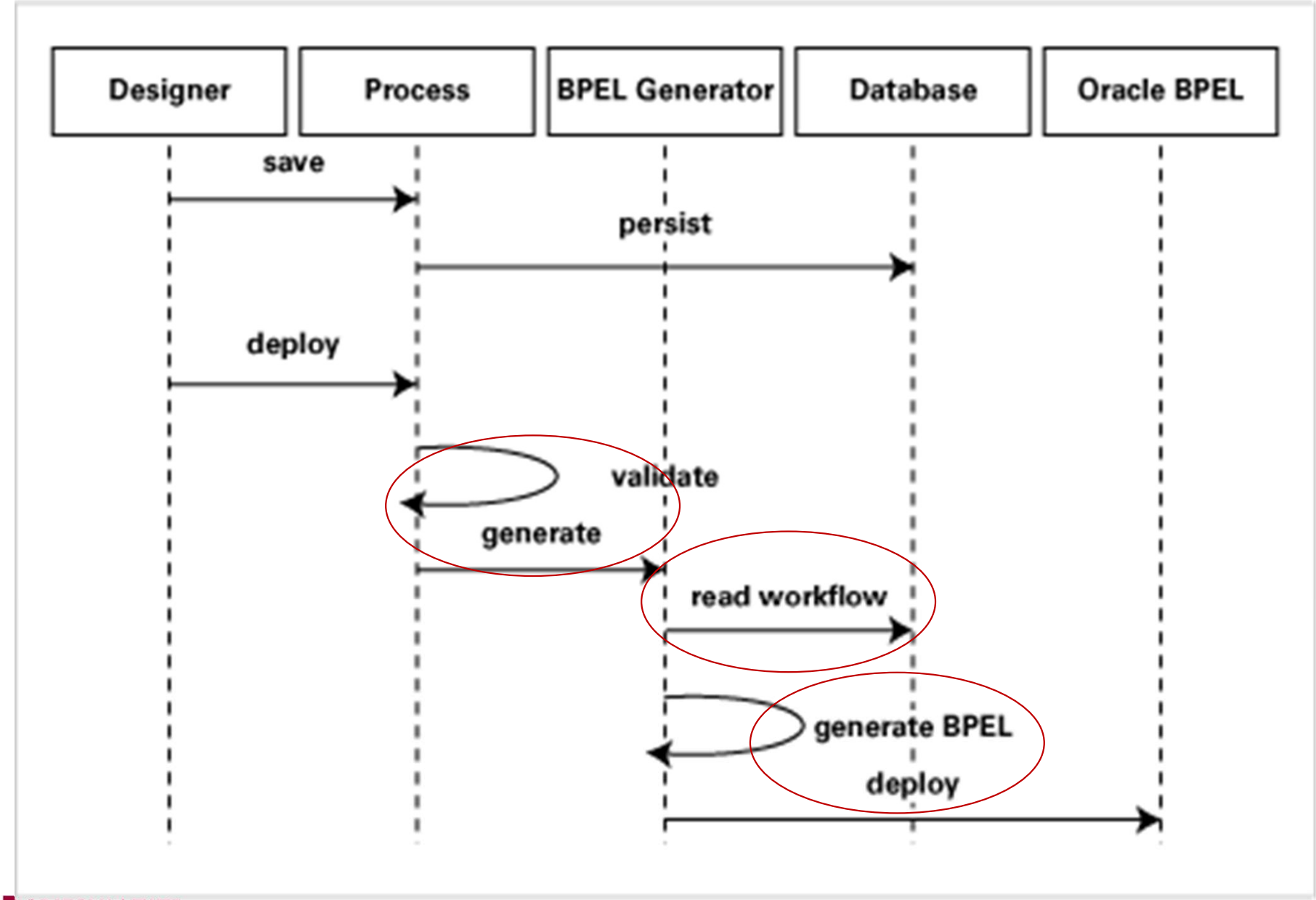


# Oracle SOA Suite for BPEL

https://docs.oracle.com/middleware/1213/core/INSOA/planning.htm#INSOA352



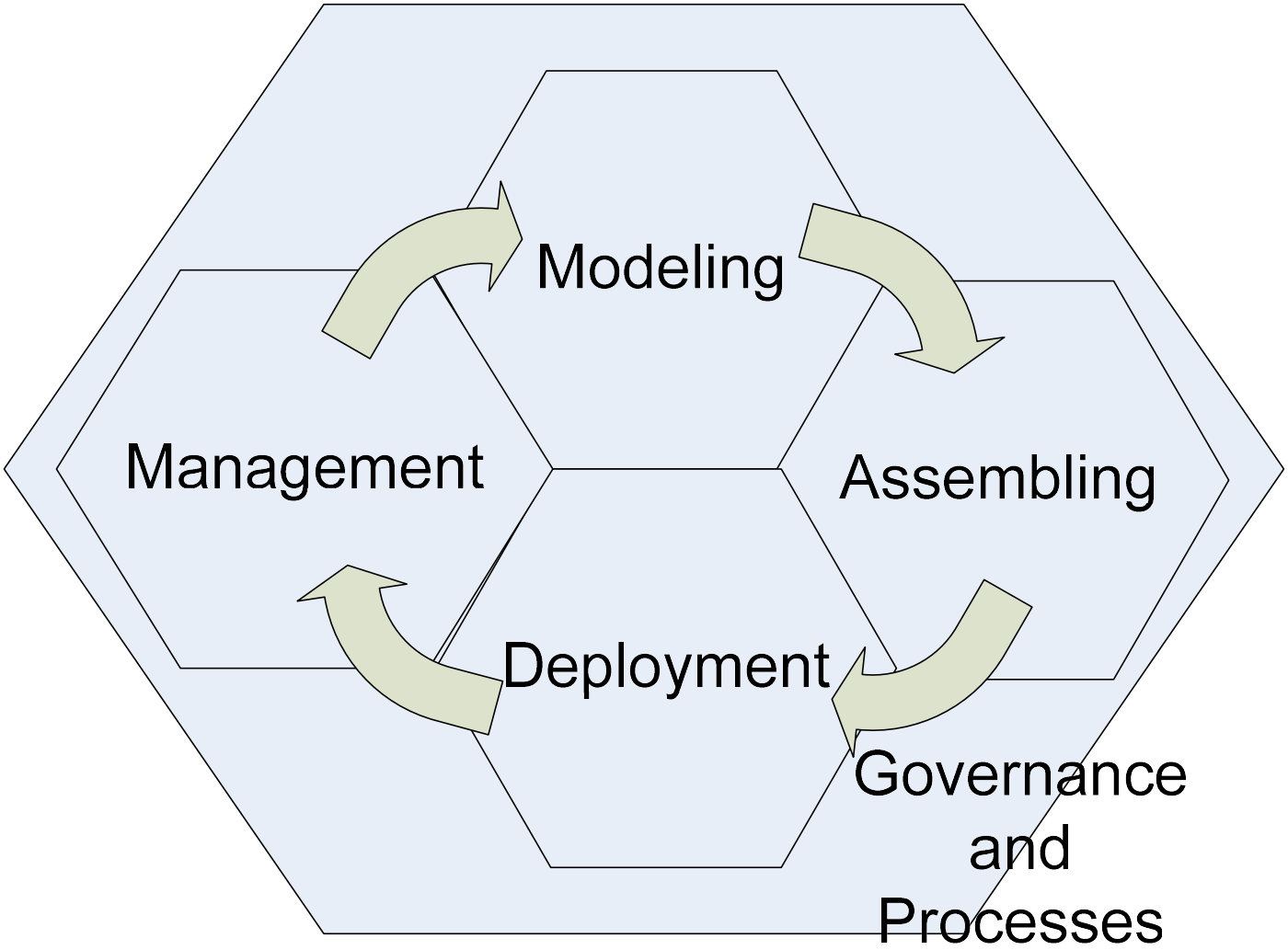
# Oracle SOA Suite for Dynamic Architecture



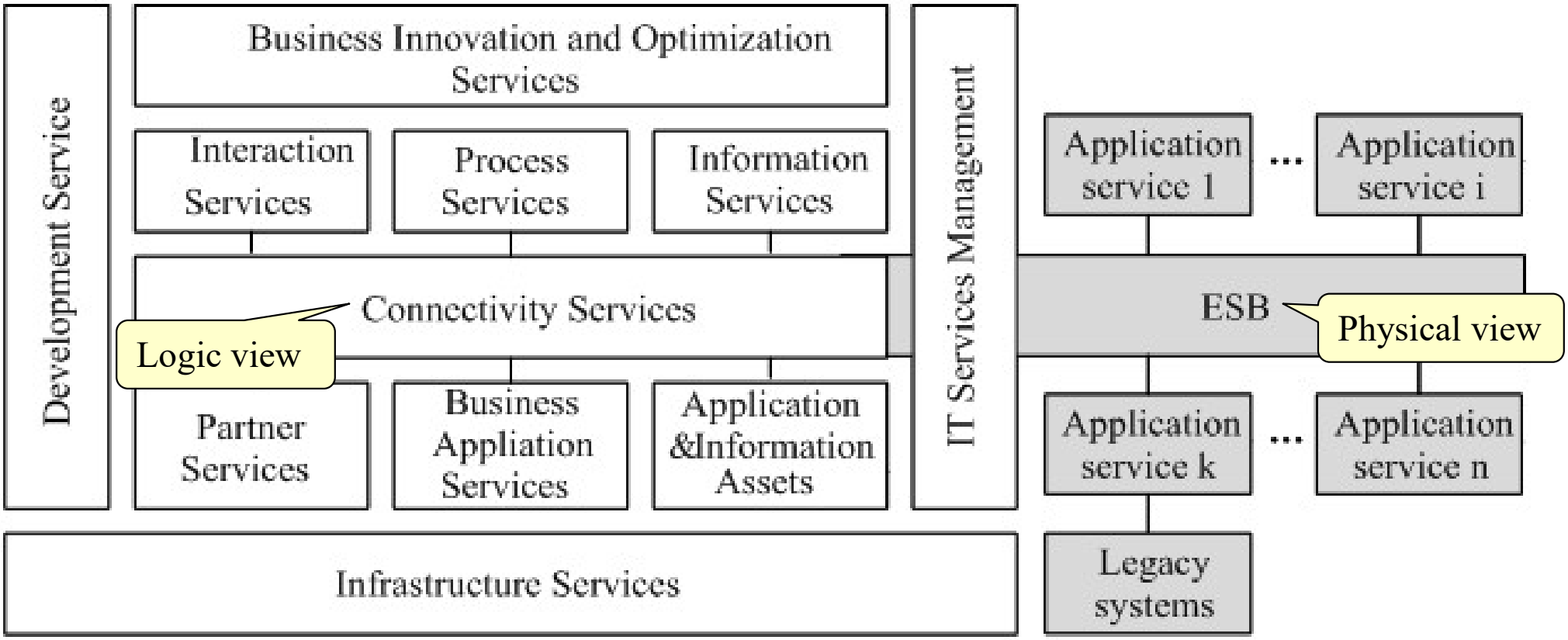
# Oracle SOA Suite for Dynamic Architecture

- An analyst uses a custom designer to graphically (visually) model the business process.
- The business process definition is stored in a database.
- The BPEL Generator reads and validates the graphical process definition, and then generates the BPEL XML file from the database representation as well as associated files used for deployment.
- The BPEL process is dynamically deployed on the Oracle BPEL Process Manager Server.
- Just-In-Time compilation can be used to convert BPEL process to executable.

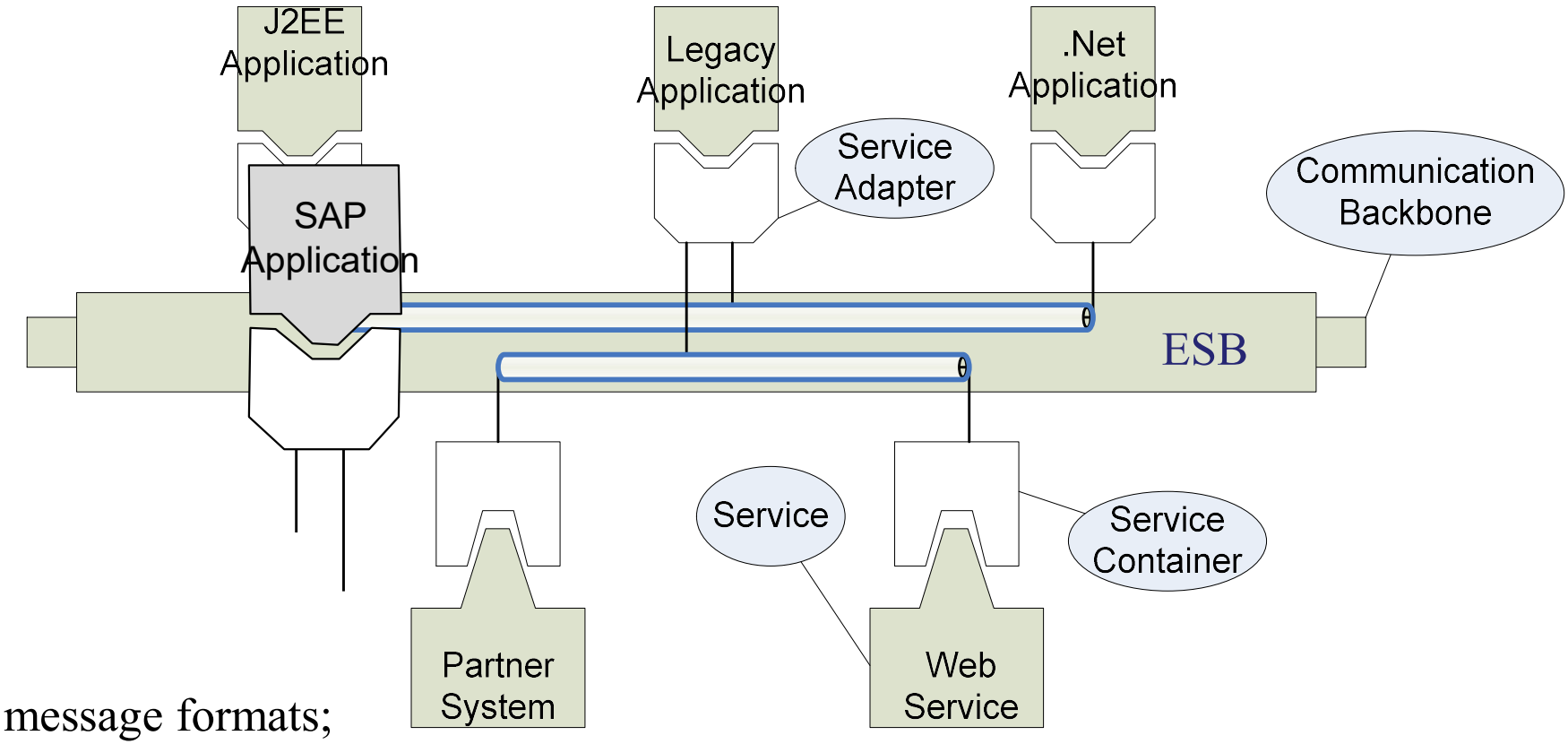
# IBM SOA Foundation Lifecycle Model (Based on Service Oriented Computing)



# IBM SOA Foundation Lifecycle Model with its Application Architecture and ESB



# IBM Enterprise Service Bus (ESB)




- Translate message formats;
- Adapt different interaction patterns;
- Convert transport protocols;
- Analyze and share the service interface;
- Propagate the security content;
- Provide WS-* support, including WS-RM, WS-Security, and WS-Transaction;
- Track and monitor service activities

# What Next?

## From Dynamic to On Demand Business

- Dynamic Business Composition
  - Changing environment and changing partners
  - Reconfiguring business without stopping operations
  - Manual reconfiguration
- On Demand Business with Artificial Intelligence
  - Proactive discovery
  - Responsive reconfiguration in real-time
  - Resilient around the world and around the clock
  - Automated reconfiguration

*On Demand eBusiness with AI-based real-time reconfig-uration, resilient & automated evolution*

*Web 5.0* with proactive artificial intelligent services and applications*

2020 -